Diffusion Models

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Abstract

1 Introduction

2 Main Results

Let $\{\varepsilon_t\}_{t=1}^T$ be a sequence of i.i.d. random variables in $\mathbb{R}^d$ with mean μ and covariance $\mathbf{I}$, where $\varepsilon_1 \sim P_{\mathcal{E}}$ belongs to the exponential family. Define

$$x_t = \sqrt{1-\beta_t}x_{t-1} + \sqrt{\beta_t}(\varepsilon_t - \mu), \tag{1}$$

Then, from (1) we have

$$q(x_t|x_{t-1}) = \frac{1}{(\sqrt{\beta_t})^d} P_{\mathcal{E}}\left(\frac{x_t - \sqrt{1-\beta_t}x_{t-1} + \sqrt{\beta_t}\mu}{\sqrt{\beta_t}}\right) \tag{2}$$

Then, we choose the exponential family such that

$$q(x_t|x_{t-1}) = \frac{1}{(\sqrt{\beta_t})^d} P_{\mathcal{E}}\left(\frac{x_t - \sqrt{1-\beta_t}x_{t-1} + \sqrt{\beta}_t\mu}{\sqrt{\beta_t}}\right) \tag{3}$$

$$= g(x_t, x_{t-1})e^{\eta_0(x_t)^T T_0(x_{t-1}) - A_0(\eta_0(x_t))}, \tag{4}$$

where $g(x_t, x_{t-1})$ is not a function of $\eta_0(x_t)$, and $A_0(\eta(x_t))$ is a function of $\eta(x_t))$. Now, from (1) we have

$$x_t = \sqrt{1-\beta_t}x_{t-1} + \sqrt{\beta_t}(\varepsilon_t - \mu) \tag{5}$$

$$= \sqrt{1-\beta_t}\sqrt{1-\beta_{t-1}}x_{t-2} + \sqrt{\beta_{t-1}}(\varepsilon_{t-1} - \mu) + \sqrt{\beta_t}(\varepsilon_t - \mu) \tag{6}$$

$$\vdots \tag{7}$$

$$= \cdots \tag{8}$$

$$\vdots \tag{9}$$

$$= \sqrt{(1-\beta_t)(1-\beta_{t-1})\cdots(1-\beta_1)}x_0 + V_t, \tag{10}$$

where

$$V_t = \sqrt{\beta_t}(\varepsilon_t - \mu) + \sqrt{(1-\beta_t)\beta_{t-1}}(\varepsilon_{t-1} - \mu) + \sqrt{(1-\beta_t)(1-\beta_{t-1})\beta_{t-2}}(\varepsilon_{t-2} - \mu) + \\ + \cdots + \sqrt{(1-\beta_t)(1-\beta_{t-1})\cdots(1-\beta_2)\beta_1}(\varepsilon_1 - \mu). \tag{11}$$

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It is easy to see that

$$\mathbb{E}[V_t] = 0, \tag{12}$$

$$\begin{aligned}\text{cov}(V_t) &= [\beta_t + (1-\beta_t)\beta_{t-1} + \cdots + (1-\beta_t)(1-\beta_{t-1})\cdots(1-\beta_2)\beta_1 \\ &= 1 - (1-\beta_1)(1-\beta_2)\cdots(1-\beta_t)]\mathbf{I},\end{aligned} \tag{13}$$

where (13) can be proved by using the induction. Now, let $\alpha_t := 1 - \beta_t$ and $\bar{\alpha}_t := \prod_{s=1}^{t} \alpha_s$, then we have

$$\text{cov}(V_t) = (1 - \bar{\alpha}_t)\mathbf{I}. \tag{14}$$

Now, let

$$K_t = \frac{V_t}{\sqrt{1-\bar{\alpha}_t}}, \tag{15}$$

then from (10) we have

$$x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1-\bar{\alpha}_t} K_t. \tag{16}$$

It follows that

$$q(x_t|x_0) = \frac{1}{\sqrt{1-\bar{\alpha}_t}} P_{K_t}\left(\frac{x_t - \sqrt{\bar{\alpha}_t}x_0}{\sqrt{1-\bar{\alpha}_t}}\right). \tag{17}$$

Now, the following result is well-known (Edgeworth Expansion).

Lemma 1 *Let $U_1, U_2, \cdots, U_n \overset{i.i.d.}{\sim} f_U(\cdot)$ with $\mathbb{E}[U] = \kappa_1$, var$(U) = \kappa_2$, higher-order cumulants $\kappa_3, \kappa_4, \cdots$, and cumulant generating function*

$$K_U(t) = \log(\mathbb{E}[e^{tU}]) = \sum_{j=1}^{\infty} \kappa_j \frac{t^j}{j!}. \tag{18}$$

Let $X_n = \frac{\sum_{i=1}^{n} U_i - n\kappa_1}{\sqrt{n\kappa_2}}$ is the standardized sample mean, then the following holds

$$f_{X_n}(x) = \varphi(x)\left(1 + \frac{\rho_3}{6\sqrt{n}} H_3(x) + O(n^{-1/2})\right), \tag{19}$$

where

$$\rho_3 = \kappa_3/\kappa_2^{3/2}. \tag{20}$$

Then, the following result is straightforward to derive.

Lemma 2 *Let $U_1, U_2, \cdots, U_n \overset{i.i.d.}{\sim} f_U(\cdot) \in \mathbb{R}^d$ with $\mathbb{E}[U_1] = \kappa_1$, cov$(U_1) = \kappa_2\mathbf{I}$, higher-order cumulants $\kappa_3, \kappa_4, \cdots$, and cumulant generating function*

$$K_{U_{1,l}}(t) = \log(\mathbb{E}[e^{tU_{1,l}}]) = \sum_{j=1}^{\infty} \kappa_{j,l} \frac{t^j}{j!}, \qquad \forall l \in [d]. \tag{21}$$

Let $X_n = \frac{\sum_{i=1}^{n} U_i - n\kappa_1}{\sqrt{n\kappa_2}}$ is the standardized sample mean, then the following holds

$$f_{X_n}(x) = \varphi(x) \prod_{l=1}^{d} \left(1 + \frac{\rho_{3,l}}{6\sqrt{n}} H_3(x_l) + O(n^{-1/2})\right) \tag{22}$$

$$= \varphi(x)\left(1 + \frac{1}{6\sqrt{n}} \sum_{l=1}^{d} \rho_{3,l} H_3(x_l) + O(n^{-1/2})\right) \qquad \forall x \in \mathbb{R}^d, \tag{23}$$

where

$$\rho_{3,l} = \kappa_{3,l}/\kappa_2^{3/2}. \tag{24}$$

Now, by Lemma 1 we have

$$q(x_t|x_0) = \frac{1}{\sqrt{1-\bar{\alpha}_t}}\varphi\left(\frac{x_t - \sqrt{\bar{\alpha}_t}x_0}{\sqrt{1-\bar{\alpha}_t}}\right)\left(1 + \frac{\rho_3}{6\sqrt{n}}H_3\left(\frac{x_t - \sqrt{\bar{\alpha}_t}x_0}{\sqrt{1-\bar{\alpha}_t}}\right) + O(n^{-1/2})\right), \tag{25}$$

and

$$q(x_{t-1}|x_0) = \frac{1}{\sqrt{1-\bar{\alpha}_{t-1}}}\varphi\left(\frac{x_{t-1} - \sqrt{\bar{\alpha}_{t-1}}x_0}{\sqrt{1-\bar{\alpha}_{t-1}}}\right)\left(1 + \frac{\rho_3}{6\sqrt{n}}H_3\left(\frac{x_{t-1} - \sqrt{\bar{\alpha}_{t-1}}x_0}{\sqrt{1-\bar{\alpha}_{t-1}}}\right) + O(n^{-1/2})\right), \tag{26}$$

Then, we have

$$q(x_{t-1}|x_t, x_0) = q(x_t|x_{t-1})\frac{q(x_{t-1}|x_0)}{q(x_t|x_0)} \tag{27}$$

$$= q(x_t|x_{t-1})\sqrt{\frac{1-\bar{\alpha}_t}{1-\bar{\alpha}_{t-1}}}\varphi\left(\frac{x_{t-1} - \sqrt{\bar{\alpha}_{t-1}}x_0}{\sqrt{1-\bar{\alpha}_{t-1}}}\right)/\varphi\left(\frac{x_t - \sqrt{\bar{\alpha}_t}x_0}{\sqrt{1-\bar{\alpha}_t}}\right)Q(x_t, x_{t-1}, x_0) \tag{28}$$

$$= h(x_{t-1}, x_t, x_0)Q(x_t, x_{t-1}, x_0), \tag{29}$$

where

$$Q(x_t, x_{t-1}, x_0) = g(x_t, x_{t-1}) \cdot \frac{1 + \frac{1}{6\sqrt{n}}\sum_{l=1}^{d}\rho_{3,l}H_3\left(\frac{x_{t-1,l} - \sqrt{\bar{\alpha}_{t-1}}x_{0,l}}{\sqrt{1-\bar{\alpha}_{t-1}}}\right) + O(n^{-1/2})}{1 + \frac{1}{6\sqrt{n}}\sum_{l=1}^{d}\rho_{3,l}H_3\left(\frac{x_{t,l} - \sqrt{\bar{\alpha}_t}x_{0,l}}{\sqrt{1-\bar{\alpha}_t}}\right) + O(n^{-1/2})}. \tag{30}$$

Then, we have

$$\log\frac{p_\theta(x_{t-1}|x_t)}{q(x_{t-1}|x_t, x_0)} = \log\frac{p_\theta(x_{t-1}|x_t)}{h(x_{t-1}, x_t, x_0)} - \log Q(x_t, x_{t-1}, x_0). \tag{31}$$

Now, we have

$$h(x_{t-1}, x_t, x_0)$$

$$= q(x_t|x_{t-1})\sqrt{\frac{1-\bar{\alpha}_t}{1-\bar{\alpha}_{t-1}}}\varphi\left(\frac{x_{t-1} - \sqrt{\bar{\alpha}_{t-1}}x_0}{\sqrt{1-\bar{\alpha}_{t-1}}}\right)/\varphi\left(\frac{x_t - \sqrt{\bar{\alpha}_t}x_0}{\sqrt{1-\bar{\alpha}_t}}\right) \tag{32}$$

$$= g(x_t, x_{t-1})\exp\left(-\frac{\|x_{t-1}\|^2 + \bar{\alpha}_{t-1}\|x_0\|^2}{2(1-\bar{\alpha}_{t-1})}\right)e^{\eta_0(x_t)^T T_0(x_{t-1}) + \frac{\sqrt{\bar{\alpha}_{t-1}}}{1-\bar{\alpha}_{t-1}}x_0 x_{t-1}}\sqrt{\frac{1-\bar{\alpha}_t}{1-\bar{\alpha}_{t-1}}}/\varphi\left(\frac{x_t - \sqrt{\bar{\alpha}_t}x_0}{\sqrt{1-\bar{\alpha}_t}}\right) \tag{33}$$

$$= g(x_t, x_{t-1})\exp\left(-\frac{\|x_{t-1}\|^2 + \bar{\alpha}_{t-1}\|x_0\|^2}{2(1-\bar{\alpha}_{t-1})}\right)e^{\eta(x_t)^T T_0(x_{t-1}) + \frac{\sqrt{\bar{\alpha}_{t-1}}}{1-\bar{\alpha}_{t-1}}x_0 x_{t-1}} \tag{34}$$

$$= g(x_t, x_{t-1})\exp\left(-\frac{\|x_{t-1}\|^2 + \bar{\alpha}_{t-1}\|x_0\|^2}{2(1-\bar{\alpha}_{t-1})}\right)e^{\eta(x_t)^T T_0(x_{t-1}) + \frac{\sqrt{\bar{\alpha}_{t-1}}}{1-\bar{\alpha}_{t-1}}x_0 x_{t-1} - A(\eta(x_t, x_0))}e^{A(\eta(x_t, x_0))}, \tag{35}$$

where

$$A(\eta(x_t, x_0)) := \log\int_{\mathbb{R}^d}\exp\left(-\frac{\|x_{t-1}\|^2}{2(1-\bar{\alpha}_{t-1})}\right)\exp\left(-\frac{1-\beta_t}{2\beta_t}\|x_{t-1}\|^2\right)e^{\eta(x_t)^\top T_0(x_{t-1}) + \frac{\sqrt{\bar{\alpha}_{t-1}}}{1-\bar{\alpha}_{t-1}}x_0^\top x_{t-1}}\,dx_{t-1}. \tag{36}$$

$$\eta(x_t, x_0) := \left[\eta_0(x_t)^T, x_t^T, \frac{\sqrt{\bar{\alpha}_{t-1}}}{1-\bar{\alpha}_{t-1}}x_0^T\right]^T. \tag{37}$$

Now, define

$$T(x_{t-1}) := [T_0(x_{t-1})^T, 0^T, x_{t-1}^T]. \tag{38}$$

Then, we have

$$
\begin{aligned}
&h(x_{t-1}, x_t, x_0) \\
&= g(x_t, x_{t-1}) \exp\left(-\frac{\|x_{t-1}\|^2}{2(1-\bar{\alpha}_{t-1})} \right) e^{\eta(x_t,x_0)^T T(x_{t-1}) - A(\eta(x_t,x_0))} e^{A(\eta(x_t,x_0))} && (39) \\
&= q'(x_t, x_{t-1}, x_0)\, \tilde{q}(x_{t-1}|x_t, x_0)\, e^{A(\eta(x_t,x_0))}, && (40)
\end{aligned}
$$

where

$$
\begin{aligned}
q'(x_t, x_{t-1}, x_0) &:= Q(x_t, x_{t-1}, x_0) \\
&= g(x_t, x_{t-1}) \cdot \frac{1 + \frac{1}{6\sqrt{n}} \sum_{l=1}^{d} \rho_{3,l} H_3\left(\frac{x_{t-1,l} - \sqrt{\bar{\alpha}_{t-1}} x_{0,l}}{\sqrt{1-\bar{\alpha}_{t-1}}} \right) + O(n^{-1/2})}{1 + \frac{1}{6\sqrt{n}} \sum_{l=1}^{d} \rho_{3,l} H_3\left(\frac{x_{t,l} - \sqrt{\bar{\alpha}_t} x_{0,l}}{\sqrt{1-\bar{\alpha}_t}} \right) + O(n^{-1/2})}, \quad (41)
\end{aligned}
$$

and

$$
\tilde{q}(x_{t-1}|x_t, x_0) := \exp\left(-\frac{\|x_{t-1}\|^2}{2(1-\bar{\alpha}_{t-1})} \right) e^{\eta(x_t,x_0)^T T(x_{t-1}) - A(\eta(x_t,x_0))}. \quad (42)
$$

Then, $\tilde{q}(x_{t-1}|x_t, x_0)$ belongs to the exponential family given x_t, x_0, and

$$
q(x_{t-1}|x_t, x_0) = q'(x_t, x_{t-1}, x_0)\, \tilde{q}(x_{t-1}|x_t, x_0). \quad (43)
$$

Hence, we have

$$
\begin{aligned}
\log \frac{p_\theta(x_{t-1}|x_t)}{q(x_{t-1}|x_t, x_0)} &= \log \frac{p_\theta(x_{t-1}|x_t)}{q'(x_t, x_{t-1}, x_0)\, \tilde{q}(x_{t-1}|x_t, x_0)} \\
&= \log \frac{p_\theta(x_{t-1}|x_t)}{\tilde{q}(x_{t-1}|x_t, x_0)} - \log q'(x_t, x_{t-1}, x_0). \quad (44)
\end{aligned}
$$

Now, choosing

$$
p_\theta(x_{t-1}|x_t) := \exp\left(-\frac{\|x_{t-1}\|^2}{2(1-\bar{\alpha}_{t-1})} \right) e^{\eta_\theta(x_t)^T T(x_{t-1}) - A(\eta_\theta(x_t))}, \quad (45)
$$

we obtain

$$
\begin{aligned}
&D_{KL}(q(x_{t-1}|x_t, x_0) \| p_\theta(x_{t-1}|x_t)) \\
&= \sum_{x_{t-1}} q(x_{t-1}|x_t, x_0) \log \frac{q(x_{t-1}|x_t, x_0)}{p_\theta(x_{t-1}|x_t)} \\
&= \sum_{x_{t-1}} q'(x_t, x_{t-1}, x_0)\, \tilde{q}(x_{t-1}|x_t, x_0) \log \frac{q'(x_t, x_{t-1}, x_0)\, \tilde{q}(x_{t-1}|x_t, x_0)}{p_\theta(x_{t-1}|x_t)} \\
&= \sum_{x_{t-1}} q' \tilde{q} \log q' + \sum_{x_{t-1}} q' \tilde{q} \log \frac{\tilde{q}}{p_\theta}. \quad (46)
\end{aligned}
$$

Assume that $q'(x_t, x_{t-1}, x_0) \le C$ and $\log q'(x_t, x_{t-1}, x_0) \le C$. Then

$$
\sum_{x_{t-1}} q' \tilde{q} \log q' \le C, \quad (47)
$$

and

$$
\sum_{x_{t-1}} q' \tilde{q} \log \frac{\tilde{q}}{p_\theta} \le C \sum_{x_{t-1}} \tilde{q}(x_{t-1}|x_t, x_0) \log \frac{\tilde{q}(x_{t-1}|x_t, x_0)}{p_\theta(x_{t-1}|x_t)}. \quad (48)
$$

Therefore,

$$
D_{KL}(q(x_{t-1}|x_t, x_0) \| p_\theta(x_{t-1}|x_t)) \le C\, D_{KL}(\tilde{q}(x_{t-1}|x_t, x_0) \| p_\theta(x_{t-1}|x_t)) + C. \quad (49)
$$

Then, minimizing L reduces to minimizing

$$\mathbb{E}_q\Big[D_{KL}(q(x_T|x_0)\|p(x_T)) + \sum_{t>1} D_{KL}(\tilde{q}(x_{t-1}|x_t,x_0)\|p_\theta(x_{t-1}|x_t)) - \log p_\theta(x_0|x_1)\Big], \quad (50)$$

up to multiplicative and additive constants independent of θ. Observe that

$$\begin{aligned} &D_{KL}(\tilde{q}(x_{t-1}|x_t,x_0)\|p_\theta(x_{t-1}|x_t)) \\ &\quad = A(\eta_\theta(x_t)) - A(\eta(x_t,x_0)) - (\eta_\theta(x_t) - \eta(x_t,x_0))^T \nabla A(\eta(x_t,x_0)). \end{aligned} \quad (51)$$

From (37) and (38), we have

$$\eta(x_t,x_0)^T T(x_{t-1}) = \eta_0(x_t)^T x_{t-1} + \frac{\sqrt{\bar{\alpha}_{t-1}}}{1-\bar{\alpha}_{t-1}} x_0^T x_{t-1} \quad (52)$$

$$= \Big[\eta_0(x_t) + \frac{\sqrt{\bar{\alpha}_{t-1}}}{1-\bar{\alpha}_{t-1}} x_0\Big]^T x_{t-1}. \quad (53)$$

Therefore, we define

$$\tilde{\eta}(x_t,x_0) := \eta_0(x_t) + \frac{\sqrt{\bar{\alpha}_{t-1}}}{1-\bar{\alpha}_{t-1}} x_0, \quad (54)$$

so that

$$\eta(x_t,x_0)^T T(x_{t-1}) = \tilde{\eta}(x_t,x_0)^T x_{t-1}. \quad (55)$$

Next, we express $\tilde{\eta}(x_t,x_0)$ in terms of the latent variable K_t. From the forward process,

$$x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1-\bar{\alpha}_t} K_t, \quad (56)$$

which implies

$$x_0 = \frac{x_t - \sqrt{1-\bar{\alpha}_t} K_t}{\sqrt{\bar{\alpha}_t}}. \quad (57)$$

Substituting this into $\tilde{\eta}(x_t,x_0)$ yields

$$\tilde{\eta}(x_t,x_0) = \eta_0(x_t) + \frac{\sqrt{\bar{\alpha}_{t-1}}}{1-\bar{\alpha}_{t-1}} \cdot \frac{x_t - \sqrt{1-\bar{\alpha}_t} K_t}{\sqrt{\bar{\alpha}_t}} \quad (58)$$

$$= \eta_0(x_t) + \frac{\sqrt{\bar{\alpha}_{t-1}}}{(1-\bar{\alpha}_{t-1})\sqrt{\bar{\alpha}_t}} x_t - \frac{\sqrt{\bar{\alpha}_{t-1}}\sqrt{1-\bar{\alpha}_t}}{(1-\bar{\alpha}_{t-1})\sqrt{\bar{\alpha}_t}} K_t. \quad (59)$$

Grouping terms, we obtain

$$\tilde{\eta}(x_t,x_0) = \tilde{\eta}_x(x_t) - b_t K_t, \quad (60)$$

where

$$\tilde{\eta}_x(x_t) := \eta_0(x_t) + \frac{\sqrt{\bar{\alpha}_{t-1}}}{(1-\bar{\alpha}_{t-1})\sqrt{\bar{\alpha}_t}} x_t, \quad (61)$$

$$b_t := \frac{\sqrt{\bar{\alpha}_{t-1}}\sqrt{1-\bar{\alpha}_t}}{(1-\bar{\alpha}_{t-1})\sqrt{\bar{\alpha}_t}}. \quad (62)$$

From (51), we have

$$\begin{aligned} D_{KL}(\tilde{q}(x_{t-1}|x_t,x_0)\|p_\theta(x_{t-1}|x_t)) \; &= A\big(\tilde{\eta}_x(x_t) - b_t K_\theta(x_t,t)\big) - A\big(\tilde{\eta}_x(x_t) - b_t K_t\big) \\ &\quad - \Big[\big(\tilde{\eta}_x(x_t) - b_t K_\theta(x_t,t)\big) - \big(\tilde{\eta}_x(x_t) - b_t K_t\big)\Big]^T \nabla A\big(\tilde{\eta}_x(x_t) - b_t K_t\big). \end{aligned} \quad (63)$$

Boundedness of the correction term q'_D

In this subsection, we distinguish between the Gaussian case and the non-Gaussian cases. Let

$$\mathcal{D}_{\mathrm{NG}}$$

denote the collection of all non-Gaussian distributions considered in this paper.

Gaussian case. For the Gaussian distribution $D = \mathrm{G}$, the transition density already belongs exactly to the exponential family. Hence, no correction term is needed, and we may write

$$q_{\mathrm{G}}(x_{t-1} \mid x_t, x_0) = \tilde{q}_{\mathrm{G}}(x_{t-1} \mid x_t, x_0), \qquad q'_{\mathrm{G}}(x_t, x_{t-1}, x_0) \equiv 1. \tag{64}$$

Therefore,

$$q'_{\mathrm{G}}(x_t, x_{t-1}, x_0) \le 1, \qquad \log q'_{\mathrm{G}}(x_t, x_{t-1}, x_0) = 0. \tag{65}$$

Non-Gaussian case. For each $D \in \mathcal{D}_{\mathrm{NG}}$, we write

$$q_D(x_{t-1} \mid x_t, x_0) = q'_D(x_t, x_{t-1}, x_0)\, \tilde{q}_D(x_{t-1} \mid x_t, x_0), \tag{66}$$

where

$$q'_D(x_t, x_{t-1}, x_0) := g_D(x_t, x_{t-1}) \cdot \frac{1 + \Delta^D_{t-1}(x_{t-1}, x_0)}{1 + \Delta^D_t(x_t, x_0)}. \tag{67}$$

Here, $g_D(x_t, x_{t-1})$ denotes the distribution-specific prefactor, and $\Delta^D_{t-1}, \Delta^D_t$ denote the corresponding Edgeworth correction terms.

Accordingly, for $D \in \mathcal{D}_{\mathrm{NG}}$, the Kullback–Leibler divergence can be decomposed as

$$\begin{aligned}
&D_{KL}(q_D(x_{t-1} \mid x_t, x_0) \,\|\, p_\theta(x_{t-1} \mid x_t)) \\
&\quad = \int q_D(x_{t-1} \mid x_t, x_0) \log \frac{q_D(x_{t-1} \mid x_t, x_0)}{p_\theta(x_{t-1} \mid x_t)}\, dx_{t-1} \\
&\quad = \int q'_D(x_t, x_{t-1}, x_0)\, \tilde{q}_D(x_{t-1} \mid x_t, x_0) \log \frac{q'_D(x_t, x_{t-1}, x_0)\, \tilde{q}_D(x_{t-1} \mid x_t, x_0)}{p_\theta(x_{t-1} \mid x_t)}\, dx_{t-1} \\
&\quad = \int q'_D(x_t, x_{t-1}, x_0)\, \tilde{q}_D(x_{t-1} \mid x_t, x_0) \log q'_D(x_t, x_{t-1}, x_0)\, dx_{t-1} \\
&\qquad + \int q'_D(x_t, x_{t-1}, x_0)\, \tilde{q}_D(x_{t-1} \mid x_t, x_0) \log \frac{\tilde{q}_D(x_{t-1} \mid x_t, x_0)}{p_\theta(x_{t-1} \mid x_t)}\, dx_{t-1}.
\end{aligned} \tag{68}$$

Lemma. Let $\mathcal{K}$ be a compact subset of the interior of the support of the distribution under consideration. Assume that the Edgeworth remainder is uniform on $\mathcal{K}$, i.e., there exists a constant $c_{\mathcal{K}} > 0$ such that

$$\sup_{(x_{t-1}, x_0) \in \mathcal{K}} |\Delta^D_{t-1}(x_{t-1}, x_0)| + \sup_{x_t \in \mathcal{K}} |\Delta^D_t(x_t, x_0)| \le \frac{c_{\mathcal{K}}}{\sqrt{n}}. \tag{69}$$

Then, for sufficiently large n, there exists a constant $C^{(R)}_{\mathcal{K}} < \infty$ such that

$$\frac{1 + \Delta^D_{t-1}(x_{t-1}, x_0)}{1 + \Delta^D_t(x_t, x_0)} \le C^{(R)}_{\mathcal{K}}, \qquad \log \frac{1 + \Delta^D_{t-1}(x_{t-1}, x_0)}{1 + \Delta^D_t(x_t, x_0)} \le \log C^{(R)}_{\mathcal{K}}. \tag{70}$$

Proof. Choose n large enough so that $c_{\mathcal{K}}/\sqrt{n} \le 1/2$. Then

$$\frac{1}{2} \le 1 + \Delta^D_{t-1}(x_{t-1}, x_0), \qquad 1 + \Delta^D_t(x_t, x_0) \le \frac{3}{2}. \tag{71}$$

Hence,

$$0 < \frac{1 + \Delta^D_{t-1}(x_{t-1}, x_0)}{1 + \Delta^D_t(x_t, x_0)} \le 3. \tag{72}$$

Therefore, one may take $C^{(R)}_{\mathcal{K}} := 3$, which yields (70). □

Proposition. Let $D \in \mathcal{D}_{\mathrm{NG}}$. Suppose that $g_D(x_t, x_{t-1})$ is continuous on a compact set $\mathcal{K}_D$ contained in the interior of its support. Then there exists a finite constant $C_D < \infty$ such that

$$q'_D(x_t, x_{t-1}, x_0) \le C_D, \qquad \log q'_D(x_t, x_{t-1}, x_0) \le C_D, \tag{73}$$

for all $(x_t, x_{t-1}, x_0) \in \mathcal{K}_D$.

Proof. Since g_D is continuous on the compact set $\mathcal{K}_D$, the Weierstrass theorem implies that

$$M_D := \sup_{(x_t, x_{t-1}) \in \mathcal{K}_D} g_D(x_t, x_{t-1}) < \infty. \tag{74}$$

Combining this with (70), we obtain

$$\begin{aligned} q'_D(x_t, x_{t-1}, x_0) &= g_D(x_t, x_{t-1}) \cdot \frac{1 + \Delta^D_{t-1}(x_{t-1}, x_0)}{1 + \Delta^D_t(x_t, x_0)} \\ &\leq M_D \, C^{(R)}_{\mathcal{K}_D}. \end{aligned} \tag{75}$$

Hence,

$$\log q'_D(x_t, x_{t-1}, x_0) \leq \log\big(M_D \, C^{(R)}_{\mathcal{K}_D}\big). \tag{76}$$

Therefore, it suffices to choose

$$C_D := \max\left\{M_D \, C^{(R)}_{\mathcal{K}_D}, \, \log\big(M_D \, C^{(R)}_{\mathcal{K}_D}\big)\right\} < \infty. \tag{77}$$

This proves (73). □

Corollary. For every distribution D, Gaussian or non-Gaussian, there exists a finite constant $C_D < \infty$ such that, on the corresponding compact set $\mathcal{K}_D$,

$$q'_D(x_t, x_{t-1}, x_0) \leq C_D, \qquad \log q'_D(x_t, x_{t-1}, x_0) \leq C_D. \tag{78}$$

Consequently, for $D \in \mathcal{D}_{\mathrm{NG}}$, (68) yields

$$\begin{aligned} &D_{KL}(q_D(x_{t-1} \mid x_t, x_0) \,\|\, p_\theta(x_{t-1} \mid x_t)) \\ &\quad \leq C_D + C_D \int \tilde{q}_D(x_{t-1} \mid x_t, x_0) \left| \log \frac{\tilde{q}_D(x_{t-1} \mid x_t, x_0)}{p_\theta(x_{t-1} \mid x_t)} \right| dx_{t-1}, \end{aligned} \tag{79}$$

while in the Gaussian case $q'_{\mathrm{G}} \equiv 1$, so the correction term disappears identically.

Metropolis–Hastings Sampling

At timestep t, we introduce the latent state

$$z := x_{t-1}, \tag{80}$$

and define the target distribution

$$\pi_t(z) := p_\theta(z \mid x_t, K_t). \tag{81}$$

We use a symmetric Gaussian random-walk proposal

$$r_t(z' \mid z) \propto \exp\left(-\frac{\|z' - z\|^2}{2}\right), \tag{82}$$

so that

$$r_t(z' \mid z) = r_t(z \mid z'). \tag{83}$$

Hence, the Metropolis–Hastings acceptance probability is

$$\alpha_t(z, z') = \min\left\{1, \, \frac{\pi_t(z') \, r_t(z \mid z')}{\pi_t(z) \, r_t(z' \mid z)}\right\} = \min\left\{1, \, \frac{p_\theta(z' \mid x_t, K_t)}{p_\theta(z \mid x_t, K_t)}\right\}. \tag{84}$$

Next, using

$$K_t = \frac{V_t}{\sqrt{1 - \bar{\alpha}_t}}, \tag{85}$$

the forward relation

$$x_t = \sqrt{\bar{\alpha}_t} \, x_0 + \sqrt{1 - \bar{\alpha}_t} \, K_t \tag{86}$$

implies

$$x_0 = \frac{x_t - \sqrt{1-\bar{\alpha}_t}\, K_t}{\sqrt{\bar{\alpha}_t}}. \tag{87}$$

Therefore, the natural parameter can be rewritten as

$$\eta_\theta(x_t, x_0) = \eta_\theta\left(x_t,\ \frac{x_t - \sqrt{1-\bar{\alpha}_t}\, K_t}{\sqrt{\bar{\alpha}_t}}\right). \tag{88}$$

For simplicity, we denote

$$\widetilde{\eta}_\theta(x_t, K_t) := \eta_\theta\left(x_t,\ \frac{x_t - \sqrt{1-\bar{\alpha}_t}\, K_t}{\sqrt{\bar{\alpha}_t}}\right). \tag{89}$$

Under the exponential-family parameterization, the reverse density is

$$p_\theta(z \mid x_t, K_t) = \exp\left(-\frac{\|z\|^2}{2(1-\bar{\alpha}_{t-1})}\right) \exp\left(\widetilde{\eta}_\theta(x_t, K_t)^T T(z) - A(\widetilde{\eta}_\theta(x_t, K_t))\right). \tag{90}$$

Hence,

$$\frac{p_\theta(z' \mid x_t, K_t)}{p_\theta(z \mid x_t, K_t)} = \frac{\exp\left(-\frac{\|z'\|^2}{2(1-\bar{\alpha}_{t-1})}\right) \exp\left(\widetilde{\eta}_\theta(x_t, K_t)^T T(z')\right)}{\exp\left(-\frac{\|z\|^2}{2(1-\bar{\alpha}_{t-1})}\right) \exp(\widetilde{\eta}_\theta(x_t, K_t)^T T(z))}, \tag{91}$$

since the log-partition term

$$A(\widetilde{\eta}_\theta(x_t, K_t)) \tag{92}$$

cancels in the ratio.

Therefore,

$$\alpha_t(z, z') = \min\left\{1,\ \frac{\exp\left(-\frac{\|z'\|^2}{2(1-\bar{\alpha}_{t-1})}\right) \exp\left(\widetilde{\eta}_\theta(x_t, K_t)^T T(z')\right)}{\exp\left(-\frac{\|z\|^2}{2(1-\bar{\alpha}_{t-1})}\right) \exp(\widetilde{\eta}_\theta(x_t, K_t)^T T(z))}\right\}. \tag{93}$$

Starting from an initial state $z^{(0)}$, the Metropolis–Hastings chain is generated as follows. At iteration m, propose

$$z' \sim r_t(\cdot \mid z^{(m)}), \tag{94}$$

and update

$$z^{(m+1)} = \begin{cases} z', & \text{with probability } \alpha_t(z^{(m)}, z'), \\ z^{(m)}, & \text{otherwise.} \end{cases} \tag{95}$$

After M iterations, we take

$$x_{t-1} = z^{(M)}. \tag{96}$$

This procedure is repeated sequentially for $t = T, T-1, \ldots, 1$ to generate the full reverse trajectory.

3 Some common exponential distributions

3.1 Gaussian Distribution

Let $\varepsilon_1, \varepsilon_2, \cdots, \varepsilon_T$ be a sequence of i.i.d. random vectors with mean $\mu = 0$ and covariance $\mathbf{I}$. Then, we have

$$q(x_t|x_{t-1}) = \frac{1}{(\sqrt{\beta_t})^d} P_{\mathcal{E}}\left(\frac{x_t - \sqrt{1-\beta_t}x_{t-1} + \sqrt{\beta}_t\mu}{\sqrt{\beta_t}}\right) \tag{97}$$

$$= \frac{1}{(\sqrt{\beta_t})^d} \exp\left(-\frac{\|x_t - \sqrt{1-\beta_t}x_{t-1}\|^2}{2\beta_t}\right) \tag{98}$$

$$= \frac{1}{(\sqrt{\beta_t})^d} \exp\left(-\frac{\|x_t\|^2}{2\beta_t}\right) \exp\left(-\frac{1-\beta_t}{2\beta_t}\|x_{t-1}\|^2\right) \exp\left(-\frac{\sqrt{1-\beta_t}}{\beta_t} x_t^T x_{t-1}\right). \tag{99}$$

Hence, we can set

$$g(x_t, x_{t-1}) = \frac{1}{(\sqrt{\beta_t})^d} \exp\Big(- \frac{\|x_t\|^2}{2\beta_t}\Big) \exp\Big(- \frac{1-\beta_t}{2\beta_t}\|x_{t-1}\|^2\Big), \tag{100}$$

$$\eta_0(x_t) = \frac{\sqrt{1-\beta_t}}{\beta_t} x_t, \tag{101}$$

$$T_0(x_{t-1}) = x_{t-1}, \tag{102}$$

$$A_0(\eta_0(x_t)) = 0. \tag{103}$$

Then, from (37) and (38) we also have

$$\eta(x_t, x_0) = \left[\frac{\sqrt{1-\beta_t}}{\beta_t} x_t^T, x_t^T, \frac{\sqrt{\bar{\alpha}_{t-1}}}{1-\bar{\alpha}_{t-1}} x_0^T \right]^T, \tag{104}$$

$$T(x_{t-1}) = [x_{t-1}^T, 0^T, x_{t-1}^T]^T. \tag{105}$$

It follows that

$$\eta(x_t, x_0)^T T(x_{t-1}) = \left(\frac{\sqrt{1-\beta_t}}{\beta_t} x_t + \frac{\sqrt{\bar{\alpha}_{t-1}}}{1-\bar{\alpha}_{t-1}} x_0 \right)^T x_{t-1}. \tag{106}$$

Hence, by (36) we obtain

$$\begin{aligned} A(\eta(x_t, x_0)) &= A(\tilde{\eta}(x_t, x_0)) \\ &= \log \int_{\mathbb{R}^d} \exp\Big(-\frac{\|x_{t-1}\|^2}{2(1-\bar{\alpha}_{t-1})}\Big) \exp\Big(-\frac{1-\beta_t}{2\beta_t}\|x_{t-1}\|^2\Big) \exp\big(\tilde{\eta}(x_t, x_0)^T x_{t-1}\big)\, dx_{t-1} \\ &= \log \int_{\mathbb{R}^d} \exp\Big(- \Big[\frac{1-\beta_t}{2\beta_t} + \frac{1}{2(1-\bar{\alpha}_{t-1})}\Big] \|x_{t-1}\|^2 + \tilde{\eta}(x_t, x_0)^T x_{t-1}\Big)\, dx_{t-1} \\ &= \log \int_{\mathbb{R}^d} \exp\Bigg(- \Big[\frac{1-\beta_t}{2\beta_t} + \frac{1}{2(1-\bar{\alpha}_{t-1})}\Big] \|x_{t-1}\|^2 + \left[\frac{\sqrt{1-\beta_t}}{\beta_t} x_t + \frac{\sqrt{\bar{\alpha}_{t-1}}}{1-\bar{\alpha}_{t-1}} x_0\right]^T x_{t-1}\Bigg)\, dx_{t-1}. \end{aligned} \tag{107}$$

where

$$\tilde{\eta}(x_t, x_0) := \frac{\sqrt{1-\beta_t}}{\beta_t} x_t + \frac{\sqrt{\bar{\alpha}_{t-1}}}{1-\bar{\alpha}_{t-1}} x_0. \tag{108}$$

Now, it is know that

$$\int_{\mathbb{R}^d} \exp\Big(- \frac{\|x-\mu\|^2}{2\sigma^2}\Big) dx = (\sqrt{2\pi\sigma^2})^d \tag{109}$$

for any $\mu \in \mathbb{R}^d, \sigma \in \mathbb{R}$. Hence, from (107) we obtain

$$A(\tilde{\eta}(x_t, x_0)) = \log\Big((\sqrt{2\pi\sigma_t^2})^d \Big) + \frac{\sigma_t^2}{2} \|\tilde{\eta}(x_t, x_0)\|^2, \tag{110}$$

where

$$\sigma_t^2 = \frac{1}{\frac{\alpha_t}{\beta_t} + \frac{1}{1-\bar{\alpha}_{t-1}}} = \frac{\beta_t(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}, \tag{111}$$

$$\tilde{\eta}(x_t, x_0) = \frac{\sqrt{\alpha_t}}{\beta_t} x_t + \frac{\sqrt{\bar{\alpha}_{t-1}}}{1-\bar{\alpha}_{t-1}} x_0. \tag{112}$$

Define

$$\begin{aligned} \tilde{\mu}_t(x_t, x_0) &:= \sigma_t^2 \tilde{\eta}(x_t, x_0) \\ &= \frac{\beta_t(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t} \left[\frac{\sqrt{1-\beta_t}}{\beta_t} x_t + \frac{\sqrt{\bar{\alpha}_{t-1}}}{1-\bar{\alpha}_{t-1}} x_0 \right] \\ &= \frac{\sqrt{1-\beta_t}(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t} x_t + \frac{\beta_t \sqrt{\bar{\alpha}_{t-1}}}{1-\bar{\alpha}_t} x_0. \end{aligned} \tag{113}$$

and

$$\mu_\theta(x_t, t) := \sigma_t^2\, \eta_\theta(x_t), \qquad \sigma_t^2 = \frac{\beta_t(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}. \tag{114}$$

Then, from (51) we obtain

$$\begin{aligned} &D_{KL}(\tilde{q}(x_{t-1} \mid x_t, x_0) \,\|\, p_\theta(x_{t-1} \mid x_t)) \\ &\quad = A(\eta_\theta(x_t)) - A(\tilde{\eta}(x_t, x_0)) - (\eta_\theta(x_t) - \tilde{\eta}(x_t, x_0))^T \nabla A(\tilde{\eta}(x_t, x_0)) \qquad (115) \\ &\quad = \frac{\sigma_t^2}{2}\Big[\|\eta_\theta(x_t)\|^2 - \|\tilde{\eta}(x_t, x_0)\|^2 - 2(\eta_\theta(x_t) - \tilde{\eta}(x_t, x_0))^T \tilde{\eta}(x_t, x_0)\Big] \\ &\quad = \frac{\sigma_t^2}{2}\big\|\eta_\theta(x_t) - \tilde{\eta}(x_t, x_0)\big\|^2, \qquad (116) \end{aligned}$$

Now, since $\tilde{q}(x_{t-1}|x_t, x_0)$ and $p_\theta(x_{t-1}|x_t)$ belong to the exponential family given x_t, x_0, hence

$$\sigma_t^2\, \tilde{\eta}(x_t, x_0) = \nabla A(\tilde{\eta}(x_t, x_0)) = \tilde{\mu}_t(x_t, x_0), \tag{117}$$

$$\sigma_t^2\, \eta_\theta(x_t) = \nabla A(\eta_\theta(x_t)) = \mu_\theta(x_t, t), \tag{118}$$

where $\tilde{\mu}_t(x_t, x_0)$ is the mean of $X \sim \tilde{q}(x_{t-1}|x_t, x_0)$, and $\mu_\theta(x_t, t)$ is the mean of $X \sim p_\theta(x_{t-1}|x_t)$ given x_t, x_0. Then, from (116) we obtain

$$\begin{aligned} &D_{KL}(\tilde{q}(x_{t-1} \mid x_t, x_0) \,\|\, p_\theta(x_{t-1} \mid x_t)) \\ &\quad = \frac{\|\tilde{\mu}_t(x_t, x_0) - \mu_\theta(x_t, t)\|_2^2}{2\sigma_t^2}. \end{aligned} \tag{119}$$

where

$$\sigma_t^2 = \frac{\beta_t(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}.$$

by (111).

Next, we express $\tilde{\eta}(x_t, x_0)$ in terms of the latent variable K_t. From the forward process,

$$x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1-\bar{\alpha}_t}\, K_t, \tag{120}$$

which implies

$$x_0 = \frac{x_t - \sqrt{1-\bar{\alpha}_t}\, K_t}{\sqrt{\bar{\alpha}_t}}. \tag{121}$$

Substituting this into the definition of $\tilde{\eta}(x_t, x_0)$ gives

$$\begin{aligned} \tilde{\eta}(x_t, x_0) &= \eta_0(x_t) + \frac{\sqrt{\bar{\alpha}_{t-1}}}{1-\bar{\alpha}_{t-1}} x_0 \\ &= \eta_0(x_t) + \frac{\sqrt{\bar{\alpha}_{t-1}}}{1-\bar{\alpha}_{t-1}} \cdot \frac{x_t - \sqrt{1-\bar{\alpha}_t}\, K_t}{\sqrt{\bar{\alpha}_t}} \\ &= \eta_0(x_t) + \frac{\sqrt{\bar{\alpha}_{t-1}}}{(1-\bar{\alpha}_{t-1})\sqrt{\bar{\alpha}_t}}\, x_t - \frac{\sqrt{\bar{\alpha}_{t-1}}\sqrt{1-\bar{\alpha}_t}}{(1-\bar{\alpha}_{t-1})\sqrt{\bar{\alpha}_t}}\, K_t. \end{aligned} \tag{122}$$

Using $\bar{\alpha}_t = \alpha_t \bar{\alpha}_{t-1}$ and

$$\eta_0(x_t) = \frac{\sqrt{1-\beta_t}}{\beta_t} x_t = \frac{\sqrt{\alpha_t}}{\beta_t} x_t,$$

we further obtain

$$\begin{aligned} \tilde{\eta}(x_t, x_0) &= \left[\frac{\sqrt{\alpha_t}}{\beta_t} + \frac{1}{(1-\bar{\alpha}_{t-1})\sqrt{\alpha_t}}\right] x_t - \frac{\sqrt{1-\bar{\alpha}_t}}{(1-\bar{\alpha}_{t-1})\sqrt{\alpha_t}}\, K_t \\ &= \frac{1-\bar{\alpha}_t}{\beta_t(1-\bar{\alpha}_{t-1})\sqrt{\alpha_t}}\, x_t - \frac{\sqrt{1-\bar{\alpha}_t}}{(1-\bar{\alpha}_{t-1})\sqrt{\alpha_t}}\, K_t. \end{aligned} \tag{123}$$

Therefore, we can write

$$\tilde{\eta}(x_t, x_0) = \tilde{\eta}_x(x_t) - b_t K_t, \tag{124}$$

where

$$\tilde{\eta}_x(x_t) := \frac{1-\bar{\alpha}_t}{\beta_t(1-\bar{\alpha}_{t-1})\sqrt{\alpha_t}}\, x_t, \tag{125}$$

$$b_t := \frac{\sqrt{1-\bar{\alpha}_t}}{(1-\bar{\alpha}_{t-1})\sqrt{\alpha_t}}. \tag{126}$$

We have

$$\eta_\theta(x_t) = \tilde{\eta}_x(x_t) - b_t K_\theta(x_t, t), \tag{127}$$

Substituting $\tilde{\eta}(x_t, x_0) = \tilde{\eta}_x(x_t) - b_t K_t$ and $\eta_\theta(x_t) = \tilde{\eta}_x(x_t) - b_t K_\theta(x_t, t)$ into (116), we obtain

$$\begin{aligned} &D_{KL}(\tilde{q}(x_{t-1} \mid x_t, x_0) \,\|\, p_\theta(x_{t-1} \mid x_t)) \\ &\quad= \frac{\sigma_t^2}{2} \left\|\eta_\theta(x_t) - \tilde{\eta}(x_t, x_0)\right\|^2 \\ &\quad= \frac{\sigma_t^2}{2} \left\|\big(\tilde{\eta}_x(x_t) - b_t K_\theta(x_t, t)\big) - \big(\tilde{\eta}_x(x_t) - b_t K_t\big)\right\|^2 \\ &\quad= \frac{\sigma_t^2 b_t^2}{2} \left\|K_\theta(x_t, t) - K_t\right\|^2. \end{aligned} \tag{128}$$

Since

$$\sigma_t^2 = \frac{\beta_t(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}, \qquad b_t = \frac{\sqrt{1-\bar{\alpha}_t}}{(1-\bar{\alpha}_{t-1})\sqrt{\alpha_t}}, \tag{129}$$

it follows that

$$\sigma_t^2 b_t^2 = \frac{\beta_t}{\alpha_t(1-\bar{\alpha}_{t-1})}. \tag{130}$$

Hence,

$$\begin{aligned} &D_{KL}(\tilde{q}(x_{t-1} \mid x_t, x_0) \,\|\, p_\theta(x_{t-1} \mid x_t)) \\ &\quad= \frac{\beta_t}{2\alpha_t(1-\bar{\alpha}_{t-1})} \left\|K_\theta(x_t, t) - K_t\right\|^2. \end{aligned} \tag{131}$$

3.2 Exponential Distribution

Let $\varepsilon_1, \varepsilon_2, \cdots, \varepsilon_T$ be a sequence of i.i.d. random vectors such that each component of ε_t follows an exponential distribution with rate λ. Then,

$$\begin{aligned} P_{\mathcal{E}}(x) &= \prod_{l=1}^{d} \lambda e^{-\lambda x_l} \mathbf{1}\{x_l \geq 0\} \\ &= \lambda^d \exp(-\lambda \mathbf{1}^T x) \mathbf{1}\{x \geq 0\}, \qquad x \in \mathbb{R}^d. \end{aligned} \tag{132}$$

Hence,

$$\begin{aligned} q(x_t|x_{t-1}) &= \frac{1}{(\sqrt{\beta_t})^d} P_{\mathcal{E}}\left(\frac{x_t - \sqrt{1-\beta_t}\, x_{t-1} + \sqrt{\beta_t}\frac{1}{\lambda}\mathbf{1}}{\sqrt{\beta_t}}\right) \\ &= \frac{\lambda^d}{(\sqrt{\beta_t})^d} \exp\left(-\frac{\lambda}{\sqrt{\beta_t}}\mathbf{1}^T x_t + \lambda\sqrt{\frac{1-\beta_t}{\beta_t}}\mathbf{1}^T x_{t-1} - d\right) \\ &\quad\times \mathbf{1}\left\{x_{t-1} \leq \frac{x_t + \frac{1}{\lambda}\sqrt{\beta_t}\mathbf{1}}{\sqrt{1-\beta_t}}\right\}. \end{aligned} \tag{133}$$

Therefore, we set

$$g(x_t, x_{t-1}) = \frac{\lambda^d}{(\sqrt{\beta_t})^d} \exp\left(-\frac{\lambda}{\sqrt{\beta_t}} \mathbf{1}^T x_t - d\right) \mathbf{1}\left\{x_{t-1} \leq \frac{x_t + \frac{1}{\lambda}\sqrt{\beta_t}\mathbf{1}}{\sqrt{1-\beta_t}}\right\}, \tag{134}$$

$$\eta_0(x_t) = \lambda\sqrt{\frac{1-\beta_t}{\beta_t}}\mathbf{1} = \lambda\sqrt{\frac{\alpha_t}{\beta_t}}\mathbf{1}, \tag{135}$$

$$T_0(x_{t-1}) = x_{t-1}, \tag{136}$$

$$A_0(\eta_0(x_t)) = 0. \tag{137}$$

Then, from (37) and (38), we define

$$\tilde{\eta}(x_t, x_0) := \eta_0(x_t) + \frac{\sqrt{\bar{\alpha}_{t-1}}}{1-\bar{\alpha}_{t-1}} x_0, \tag{138}$$

so that

$$\eta(x_t, x_0)^T T(x_{t-1}) = \tilde{\eta}(x_t, x_0)^T x_{t-1}. \tag{139}$$

Hence, by (36) we obtain

$$\begin{aligned} A(\eta(x_t, x_0)) &= A(\tilde{\eta}(x_t, x_0)) \\ &= \log \int_{\mathbb{R}^d} \exp\left(-\frac{\|x_{t-1}\|^2}{2(1-\bar{\alpha}_{t-1})}\right) \exp\left(-\frac{1-\beta_t}{2\beta_t}\|x_{t-1}\|^2\right) \exp\big(\tilde{\eta}(x_t, x_0)^\top x_{t-1}\big)\, dx_{t-1} \\ &= \log \int_{\mathbb{R}^d} \exp\left(-\frac{1}{2}\left(\frac{1}{1-\bar{\alpha}_{t-1}} + \frac{1-\beta_t}{\beta_t}\right)\|x_{t-1}\|^2 + \tilde{\eta}(x_t, x_0)^\top x_{t-1}\right) dx_{t-1}. \end{aligned} \tag{140}$$

Now, define

$$\sigma_t^2 := \frac{1}{\frac{1}{1-\bar{\alpha}_{t-1}} + \frac{1-\beta_t}{\beta_t}} = \frac{\beta_t(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}. \tag{141}$$

Then (140) becomes

$$\begin{aligned} A(\tilde{\eta}(x_t, x_0)) &= \log \int_{\mathbb{R}^d} \exp\left(-\frac{1}{2\sigma_t^2}\|x_{t-1}\|^2 + \tilde{\eta}(x_t, x_0)^\top x_{t-1}\right) dx_{t-1} \\ &= \log \int_{\mathbb{R}^d} \exp\left(-\frac{1}{2\sigma_t^2}\|x_{t-1} - \sigma_t^2\tilde{\eta}(x_t, x_0)\|^2 + \frac{\sigma_t^2}{2}\|\tilde{\eta}(x_t, x_0)\|^2\right) dx_{t-1} \\ &= \log\left((\sqrt{2\pi\sigma_t^2})^d\right) + \frac{\sigma_t^2}{2}\|\tilde{\eta}(x_t, x_0)\|^2. \end{aligned} \tag{142}$$

Therefore,

$$\begin{aligned} &D_{KL}(\tilde{q}(x_{t-1} \mid x_t, x_0) \,\|\, p_\theta(x_{t-1} \mid x_t)) \\ &\quad = A(\eta_\theta(x_t, t)) - A(\tilde{\eta}(x_t, x_0)) - (\eta_\theta(x_t, t) - \tilde{\eta}(x_t, x_0))^\top \nabla A(\tilde{\eta}(x_t, x_0)) \\ &\quad = \frac{\sigma_t^2}{2}\left\|\eta_\theta(x_t, t) - \tilde{\eta}(x_t, x_0)\right\|^2, \end{aligned} \tag{143}$$

Next, we express $\tilde{\eta}(x_t, x_0)$ in terms of the latent variable K_t. From the forward process,

$$x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1-\bar{\alpha}_t}\, K_t, \tag{144}$$

which implies

$$x_0 = \frac{x_t - \sqrt{1-\bar{\alpha}_t}\, K_t}{\sqrt{\bar{\alpha}_t}}. \tag{145}$$

Substituting this into $\tilde{\eta}(x_t, x_0)$ gives

$$\begin{aligned}\tilde{\eta}(x_t, x_0) &= \eta_0(x_t) + \frac{\sqrt{\bar{\alpha}_{t-1}}}{1-\bar{\alpha}_{t-1}} \cdot \frac{x_t - \sqrt{1-\bar{\alpha}_t}\, K_t}{\sqrt{\bar{\alpha}_t}} \\ &= \eta_0(x_t) + \frac{1}{(1-\bar{\alpha}_{t-1})\sqrt{\alpha_t}}\, x_t - \frac{\sqrt{1-\bar{\alpha}_t}}{(1-\bar{\alpha}_{t-1})\sqrt{\alpha_t}}\, K_t. \end{aligned} \tag{146}$$

Hence,

$$\tilde{\eta}(x_t, x_0) = \tilde{\eta}_x(x_t) - b_t K_t, \tag{147}$$

where

$$\tilde{\eta}_x(x_t) := \lambda\sqrt{\frac{\alpha_t}{\beta_t}}\mathbf{1} + \frac{1}{(1-\bar{\alpha}_{t-1})\sqrt{\alpha_t}}\, x_t, \tag{148}$$

$$b_t := \frac{\sqrt{1-\bar{\alpha}_t}}{(1-\bar{\alpha}_{t-1})\sqrt{\alpha_t}}. \tag{149}$$

We parameterize the model natural parameter in the same form:

$$\eta_\theta(x_t, t) = \tilde{\eta}_x(x_t) - b_t K_\theta(x_t, t), \tag{150}$$

Substituting $\tilde{\eta}(x_t, x_0) = \tilde{\eta}_x(x_t) - b_t K_t$ and $\eta_\theta(x_t, t) = \tilde{\eta}_x(x_t) - b_t K_\theta(x_t, t)$ into (143), we obtain

$$\begin{aligned} &D_{KL}(\tilde{q}(x_{t-1} \mid x_t, x_0) \,\|\, p_\theta(x_{t-1} \mid x_t)) \\ &\quad= \frac{\sigma_t^2}{2}\, \|(\tilde{\eta}_x(x_t) - b_t K_\theta(x_t, t)) - (\tilde{\eta}_x(x_t) - b_t K_t)\|^2 \\ &\quad= \frac{\sigma_t^2 b_t^2}{2} \|K_\theta(x_t, t) - K_t\|^2. \end{aligned} \tag{151}$$

Since

$$\begin{aligned}\sigma_t^2 b_t^2 &= \frac{\beta_t(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t} \cdot \frac{1-\bar{\alpha}_t}{\alpha_t(1-\bar{\alpha}_{t-1})^2} \\ &= \frac{\beta_t}{\alpha_t(1-\bar{\alpha}_{t-1})}, \end{aligned} \tag{152}$$

it follows that

$$D_{KL}(\tilde{q}(x_{t-1} \mid x_t, x_0) \,\|\, p_\theta(x_{t-1} \mid x_t)) = \frac{\beta_t}{2\alpha_t(1-\bar{\alpha}_{t-1})} \|K_\theta(x_t, t) - K_t\|^2. \tag{153}$$

Therefore, a natural training objective is

$$\tilde{L} = \mathbb{E}_q\left[D_{KL}(q(x_T|x_0)\|p(x_T)) + \sum_{t>1} \frac{\beta_t}{2\alpha_t(1-\bar{\alpha}_{t-1})} \|K_\theta(x_t, t) - K_t\|^2 - \log p_\theta(x_0|x_1)\right]. \tag{154}$$

3.3 Laplace Distribution

Consider the zero-mean Laplace distribution

$$p(x) = \frac{1}{2b} \exp\left(-\frac{|x|}{b}\right), \qquad b > 0.$$

Using the Gaussian scale mixture (GSM) representation, we introduce a latent variance

$$v \sim \mathrm{Exp}\left(\frac{1}{2b^2}\right),$$

such that, conditioned on v, the Laplace variable becomes Gaussian:

$$x \mid v \sim \mathcal{N}(0, v).$$

Therefore, conditioned on v, the forward transition is Gaussian with variance $\beta_t v$:

$$q(x_t \mid x_{t-1}, v) = \frac{1}{(\sqrt{2\pi\beta_t v})^d} \exp\left(-\frac{\|x_t - \sqrt{1-\beta_t}\, x_{t-1}\|^2}{2\beta_t v}\right). \tag{155}$$

Expanding the quadratic term gives

$$q(x_t \mid x_{t-1}, v) = \frac{1}{(\sqrt{2\pi\beta_t v})^d} \exp\left(-\frac{\|x_t\|^2}{2\beta_t v}\right) \exp\left(-\frac{1-\beta_t}{2\beta_t v}\|x_{t-1}\|^2\right) \exp\left(\frac{\sqrt{1-\beta_t}}{\beta_t v} x_t^T x_{t-1}\right). \tag{156}$$

Hence, we can set

$$g(x_t, x_{t-1}, v) = \frac{1}{(\sqrt{2\pi\beta_t v})^d} \exp\left(-\frac{\|x_t\|^2}{2\beta_t v}\right) \exp\left(-\frac{1-\beta_t}{2\beta_t v}\|x_{t-1}\|^2\right), \tag{157}$$

$$\eta_0(x_t, v) = \frac{\sqrt{1-\beta_t}}{\beta_t v}\, x_t, \tag{158}$$

$$T_0(x_{t-1}) = x_{t-1}, \tag{159}$$

$$A_0(\eta_0(x_t, v)) = 0. \tag{160}$$

We define

$$\tilde{\eta}(x_t, x_0, v) := \eta_0(x_t, v) + \frac{\sqrt{\bar{\alpha}_{t-1}}}{1-\bar{\alpha}_{t-1}} x_0 = \frac{\sqrt{1-\beta_t}}{\beta_t v} x_t + \frac{\sqrt{\bar{\alpha}_{t-1}}}{1-\bar{\alpha}_{t-1}} x_0, \tag{161}$$

so that

$$\eta(x_t, x_0, v)^T T(x_{t-1}) = \tilde{\eta}(x_t, x_0, v)^T x_{t-1}. \tag{162}$$

Hence, by (36) we obtain

$$A(\tilde{\eta}(x_t, x_0, v)) = \log \int_{\mathbb{R}^d} \exp\left(-\left[\frac{1-\beta_t}{2\beta_t v} + \frac{1}{2(1-\bar{\alpha}_{t-1})}\right]\|x_{t-1}\|^2 + \tilde{\eta}(x_t, x_0, v)^T x_{t-1}\right) dx_{t-1}. \tag{163}$$

Using the Gaussian integral identity, we obtain

$$A(\tilde{\eta}(x_t, x_0, v)) = \log\left((\sqrt{2\pi\sigma_{t,v}^2})^d\right) + \frac{\sigma_{t,v}^2}{2}\|\tilde{\eta}(x_t, x_0, v)\|^2, \tag{164}$$

where

$$\sigma_{t,v}^2 = \frac{1}{\frac{1-\beta_t}{\beta_t v} + \frac{1}{1-\bar{\alpha}_{t-1}}} = \frac{\beta_t v(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t + (v-1)(1-\beta_t)(1-\bar{\alpha}_{t-1})}. \tag{165}$$

Define

$$\tilde{\mu}_t(x_t, x_0, v) := \sigma_{t,v}^2\, \tilde{\eta}(x_t, x_0, v), \tag{166}$$

$$\mu_\theta(x_t, t, v) := \sigma_{t,v}^2\, \eta_\theta(x_t, v). \tag{167}$$

Then, from (51), we obtain

$$\begin{aligned} &D_{KL}(\tilde{q}(x_{t-1} \mid x_t, x_0, v) \,\|\, p_\theta(x_{t-1} \mid x_t, v)) \\ &\quad = A(\eta_\theta(x_t, v)) - A(\tilde{\eta}(x_t, x_0, v)) - (\eta_\theta(x_t, v) - \tilde{\eta}(x_t, x_0, v))^T \nabla A(\tilde{\eta}(x_t, x_0, v)) \\ &\quad = \frac{\sigma_{t,v}^2}{2}\|\eta_\theta(x_t, v) - \tilde{\eta}(x_t, x_0, v)\|^2 \qquad (168) \\ &\quad = \frac{\|\tilde{\mu}_t(x_t, x_0, v) - \mu_\theta(x_t, t, v)\|^2}{2\sigma_{t,v}^2}. \qquad (169) \end{aligned}$$

Next, we express $\tilde{\eta}(x_t, x_0, v)$ in terms of the latent variable K_t. From the forward process,

$$x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1-\bar{\alpha}_t}\, K_t, \tag{170}$$

which implies

$$x_0 = \frac{x_t - \sqrt{1-\bar{\alpha}_t}\, K_t}{\sqrt{\bar{\alpha}_t}}. \tag{171}$$

Substituting this into (161) yields

$$\begin{aligned}\tilde{\eta}(x_t, x_0, v) &= \eta_0(x_t, v) + \frac{\sqrt{\bar{\alpha}_{t-1}}}{1-\bar{\alpha}_{t-1}} \cdot \frac{x_t - \sqrt{1-\bar{\alpha}_t}\, K_t}{\sqrt{\bar{\alpha}_t}} \\ &= \eta_0(x_t, v) + \frac{1}{(1-\bar{\alpha}_{t-1})\sqrt{\alpha_t}}\, x_t - \frac{\sqrt{1-\bar{\alpha}_t}}{(1-\bar{\alpha}_{t-1})\sqrt{\alpha_t}}\, K_t. \end{aligned} \tag{172}$$

Therefore, we can write

$$\tilde{\eta}(x_t, x_0, v) = \tilde{\eta}_x(x_t, v) - b_t K_t, \tag{173}$$

where

$$\tilde{\eta}_x(x_t, v) := \frac{\sqrt{1-\beta_t}}{\beta_t v} x_t + \frac{1}{(1-\bar{\alpha}_{t-1})\sqrt{\alpha_t}}\, x_t, \tag{174}$$

$$b_t := \frac{\sqrt{1-\bar{\alpha}_t}}{(1-\bar{\alpha}_{t-1})\sqrt{\alpha_t}}. \tag{175}$$

We parameterize the model natural parameter in the same form:

$$\eta_\theta(x_t, v) = \tilde{\eta}_x(x_t, v) - b_t K_\theta(x_t, t, v). \tag{176}$$

Substituting this and $\tilde{\eta}(x_t, x_0, v) = \tilde{\eta}_x(x_t, v) - b_t K_t$ into (168), we obtain

$$\begin{aligned} &D_{KL}(\tilde{q}(x_{t-1} \mid x_t, x_0, v) \,\|\, p_\theta(x_{t-1} \mid x_t, v)) \\ &= \frac{\sigma_{t,v}^2 b_t^2}{2} \|K_\theta(x_t, t, v) - K_t\|^2. \end{aligned} \tag{177}$$

Therefore, conditioned on v, the denoising objective can be written as

$$\mathcal{L}_t^{\mathrm{GSM}}(\theta) \propto \mathbb{E}_{x_0, x_t, v}\left[\frac{\sigma_{t,v}^2 b_t^2}{2} \|K_\theta(x_t, t, v) - K_t\|^2\right]. \tag{178}$$

3.4 Gamma Distribution

Let $\varepsilon_1, \varepsilon_2, \ldots, \varepsilon_T$ be a sequence of i.i.d. random vectors, where each component follows a Gamma distribution with shape $k > 0$ and rate $\lambda > 0$. Let

$$\mu = \frac{k}{\lambda},$$

and define the forward process

$$x_t = \sqrt{\alpha_t}\, x_{t-1} + \sqrt{\beta_t}\,(\varepsilon_t - \mu), \qquad \alpha_t = 1 - \beta_t. \tag{179}$$

Then,

$$q(x_t \mid x_{t-1}) = \frac{1}{(\sqrt{\beta_t})^d} P_\varepsilon\left(\frac{x_t - \sqrt{\alpha_t} x_{t-1} + \sqrt{\beta_t}\mu}{\sqrt{\beta_t}}\right) \tag{180}$$

where

$$P_\varepsilon(z) = \prod_{i=1}^{d} \frac{\lambda^k}{\Gamma(k)} z_i^{k-1} e^{-\lambda z_i} \mathbf{1}_{\{z_i > 0\}}. \tag{181}$$

Ignoring the global normalizing constant depending only on (k, λ), we obtain

$$q(x_t \mid x_{t-1}) \propto \frac{1}{(\sqrt{\beta_t})^d} \prod_{i=1}^{d} z_{t,i}^{k-1} \exp(-\lambda z_{t,i}) \mathbf{1}_{\{z_{t,i} > 0\}} \tag{182}$$

where

$$z_t = \frac{x_t - \sqrt{\alpha_t} x_{t-1} + \sqrt{\beta_t}\mu}{\sqrt{\beta_t}} = a_t - b_t x_{t-1}, \tag{183}$$

with

$$a_t = \frac{x_t + \sqrt{\beta_t}\mu}{\sqrt{\beta_t}}, \qquad b_t = \sqrt{\frac{\alpha_t}{\beta_t}}. \tag{184}$$

Hence,

$$q(x_t \mid x_{t-1}) = g(x_t, x_{t-1}) \exp\big(\eta_0(x_t)^T T_0(x_{t-1})\big) \tag{185}$$

where

$$g(x_t, x_{t-1}) = \frac{1}{(\sqrt{\beta_t})^d} \prod_{i=1}^{d} (a_{t,i} - b_t x_{t-1,i})^{k-1} \exp(-\lambda a_{t,i}) \mathbf{1}_{\{x_{t-1,i} \le c_{t,i}\}} \tag{186}$$

$$\eta_0(x_t) = \lambda b_t \, \mathbf{1} = \lambda \sqrt{\frac{\alpha_t}{\beta_t}} \, \mathbf{1} \tag{187}$$

$$T_0(x_{t-1}) = x_{t-1} \tag{188}$$

$$A_0(\eta_0(x_t)) = 0. \tag{189}$$

Then, we define

$$\tilde{\eta}(x_t, x_0) := \eta_0(x_t) + \frac{\sqrt{\bar{\alpha}_{t-1}}}{1 - \bar{\alpha}_{t-1}} x_0 = \lambda \sqrt{\frac{\alpha_t}{\beta_t}} \, \mathbf{1} + \frac{\sqrt{\bar{\alpha}_{t-1}}}{1 - \bar{\alpha}_{t-1}} x_0 \tag{190}$$

so that

$$\eta(x_t, x_0)^T T(x_{t-1}) = \tilde{\eta}(x_t, x_0)^T x_{t-1}. \tag{191}$$

For the Gamma distribution, we use the same log-partition form as above. Hence,

$$\begin{aligned} A(\eta(x_t, x_0)) &= A(\tilde{\eta}(x_t, x_0)) \\ &= \log \int_{\mathbb{R}^d} \exp\left(-\frac{\|x_{t-1}\|^2}{2(1 - \bar{\alpha}_{t-1})}\right) \exp\left(-\frac{1 - \beta_t}{2\beta_t} \|x_{t-1}\|^2\right) \exp\big(\tilde{\eta}(x_t, x_0)^T x_{t-1}\big) \, dx_{t-1}. \end{aligned} \tag{192}$$

Equivalently,

$$A(\tilde{\eta}(x_t, x_0)) = \log \int_{\mathbb{R}^d} \exp\left(-\frac{1}{2}\left(\frac{1}{1 - \bar{\alpha}_{t-1}} + \frac{1 - \beta_t}{\beta_t}\right) \|x_{t-1}\|^2 + \tilde{\eta}(x_t, x_0)^T x_{t-1}\right) \, dx_{t-1}. \tag{193}$$

Define

$$\sigma_t^2 := \frac{1}{\frac{1}{1 - \bar{\alpha}_{t-1}} + \frac{1 - \beta_t}{\beta_t}} = \frac{\beta_t (1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t}. \tag{194}$$

Then, using the Gaussian integral identity, we obtain

$$A(\tilde{\eta}(x_t, x_0)) = \log\left((\sqrt{2\pi\sigma_t^2})^d\right) + \frac{\sigma_t^2}{2} \|\tilde{\eta}(x_t, x_0)\|^2. \tag{195}$$

Therefore,

$$\begin{aligned} &D_{KL}(\tilde{q}(x_{t-1} \mid x_t, x_0) \,\|\, p_\theta(x_{t-1} \mid x_t)) \\ &\quad = A(\eta_\theta(x_t, t)) - A(\tilde{\eta}(x_t, x_0)) - (\eta_\theta(x_t, t) - \tilde{\eta}(x_t, x_0))^T \nabla A(\tilde{\eta}(x_t, x_0)) \\ &\quad = \frac{\sigma_t^2}{2} \left\|\eta_\theta(x_t, t) - \tilde{\eta}(x_t, x_0)\right\|^2. \end{aligned} \tag{196}$$

Next, we express $\tilde{\eta}(x_t, x_0)$ in terms of the latent variable K_t. From the forward process,

$$x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1-\bar{\alpha}_t}\, K_t \tag{197}$$

which implies

$$x_0 = \frac{x_t - \sqrt{1-\bar{\alpha}_t}\, K_t}{\sqrt{\bar{\alpha}_t}}. \tag{198}$$

Substituting this into $\tilde{\eta}(x_t, x_0)$ gives

$$\begin{aligned}\tilde{\eta}(x_t, x_0) &= \eta_0(x_t) + \frac{\sqrt{\bar{\alpha}_{t-1}}}{1-\bar{\alpha}_{t-1}} \cdot \frac{x_t - \sqrt{1-\bar{\alpha}_t}\, K_t}{\sqrt{\bar{\alpha}_t}} \\ &= \eta_0(x_t) + \frac{1}{(1-\bar{\alpha}_{t-1})\sqrt{\alpha_t}}\, x_t - \frac{\sqrt{1-\bar{\alpha}_t}}{(1-\bar{\alpha}_{t-1})\sqrt{\alpha_t}}\, K_t. \end{aligned} \tag{199}$$

Therefore, we can write

$$\tilde{\eta}(x_t, x_0) = \tilde{\eta}_x(x_t) - b_t K_t \tag{200}$$

where

$$\tilde{\eta}_x(x_t) := \eta_0(x_t) + \frac{1}{(1-\bar{\alpha}_{t-1})\sqrt{\alpha_t}}\, x_t, \tag{201}$$

$$b_t := \frac{\sqrt{1-\bar{\alpha}_t}}{(1-\bar{\alpha}_{t-1})\sqrt{\alpha_t}}. \tag{202}$$

We parameterize the model natural parameter in the same form:

$$\eta_\theta(x_t, t) = \tilde{\eta}_x(x_t) - b_t K_\theta(x_t, t), \tag{203}$$

Substituting $\tilde{\eta}(x_t, x_0) = \tilde{\eta}_x(x_t) - b_t K_t$ and $\eta_\theta(x_t, t) = \tilde{\eta}_x(x_t) - b_t K_\theta(x_t, t)$ into (196), we obtain

$$\begin{aligned} &D_{KL}(\tilde{q}(x_{t-1} \mid x_t, x_0) \,\|\, p_\theta(x_{t-1} \mid x_t)) \\ &\quad = \frac{\sigma_t^2}{2} \left\| (\tilde{\eta}_x(x_t) - b_t K_\theta(x_t, t)) - (\tilde{\eta}_x(x_t) - b_t K_t) \right\|^2 \\ &\quad = \frac{\sigma_t^2 b_t^2}{2} \|K_\theta(x_t, t) - K_t\|^2. \end{aligned} \tag{204}$$

Since

$$\begin{aligned} \sigma_t^2 b_t^2 &= \frac{\beta_t(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t} \cdot \frac{1-\bar{\alpha}_t}{\alpha_t(1-\bar{\alpha}_{t-1})^2} \\ &= \frac{\beta_t}{\alpha_t(1-\bar{\alpha}_{t-1})}, \end{aligned} \tag{205}$$

it follows that

$$D_{KL}(\tilde{q}(x_{t-1} \mid x_t, x_0) \,\|\, p_\theta(x_{t-1} \mid x_t)) = \frac{\beta_t}{2\alpha_t(1-\bar{\alpha}_{t-1})} \|K_\theta(x_t, t) - K_t\|^2. \tag{206}$$

Therefore, a natural training objective is

$$\tilde{L} = \mathbb{E}_q \left[D_{KL}(q(x_T \mid x_0) \| p(x_T)) + \sum_{t>1} \frac{\beta_t}{2\alpha_t(1-\bar{\alpha}_{t-1})} \|K_\theta(x_t, t) - K_t\|^2 - \log p_\theta(x_0 \mid x_1) \right].$$

3.5 Beta Distribution

Let $\varepsilon_1, \varepsilon_2, \ldots, \varepsilon_T$ be a sequence of i.i.d. random vectors such that each coordinate follows $\mathrm{Beta}(\alpha, \beta)$, applied component-wise, with mean

$$\mu = \frac{\alpha}{\alpha + \beta} \in (0, 1).$$

For a fixed variance schedule $(\beta_t)_{t=1}^T$ with $\alpha_t = 1 - \beta_t$, we define the forward process

$$x_t = \sqrt{\alpha_t}\, x_{t-1} + \sqrt{\beta_t}\,(\varepsilon_t - \mu\mathbf{1}). \tag{207}$$

Then,

$$q(x_t \mid x_{t-1}) = \frac{1}{(\sqrt{\beta_t})^d} P_\varepsilon\left(\frac{x_t - \sqrt{\alpha_t}x_{t-1} + \sqrt{\beta_t}\mu\mathbf{1}}{\sqrt{\beta_t}}\right), \tag{208}$$

where

$$P_\varepsilon(z) \propto \prod_{i=1}^d z_i^{\alpha-1}(1-z_i)^{\beta-1}\mathbf{1}_{\{0<z_i<1\}}. \tag{209}$$

Hence,

$$q(x_t \mid x_{t-1}) \propto \frac{1}{(\sqrt{\beta_t})^d} \prod_{i=1}^d z_{t,i}^{\alpha-1}(1-z_{t,i})^{\beta-1}\mathbf{1}_{\{0<z_{t,i}<1\}}, \tag{210}$$

where

$$z_t = \frac{x_t - \sqrt{\alpha_t}x_{t-1} + \sqrt{\beta_t}\mu\mathbf{1}}{\sqrt{\beta_t}} = a_t - b_t x_{t-1}, \tag{211}$$

with

$$a_t = \frac{x_t + \sqrt{\beta_t}\mu\mathbf{1}}{\sqrt{\beta_t}}, \qquad b_t = \sqrt{\frac{\alpha_t}{\beta_t}}. \tag{212}$$

Equivalently,

$$q(x_t \mid x_{t-1}) \propto \frac{1}{(\sqrt{\beta_t})^d} \prod_{i=1}^d (a_{t,i} - b_t x_{t-1,i})^{\alpha-1}(1 - a_{t,i} + b_t x_{t-1,i})^{\beta-1}\mathbf{1}_{\{\ell_{t,i}<x_{t-1,i}<u_{t,i}\}}, \tag{213}$$

where

$$\ell_{t,i} = \frac{a_{t,i} - 1}{b_t}, \qquad u_{t,i} = \frac{a_{t,i}}{b_t}. \tag{214}$$

We write

$$q(x_t \mid x_{t-1}) = g_B(x_t, x_{t-1}) \exp\big(\eta_0(x_t)^T T_0(x_{t-1})\big), \tag{215}$$

where

$$g_B(x_t, x_{t-1}) = \frac{1}{(\sqrt{\beta_t})^d} \prod_{i=1}^d (a_{t,i} - b_t x_{t-1,i})^{\alpha-1}(1 - a_{t,i} + b_t x_{t-1,i})^{\beta-1}\mathbf{1}_{\{\ell_{t,i}<x_{t-1,i}<u_{t,i}\}}, \tag{216}$$

$$\eta_0(x_t) = 0, \tag{217}$$

$$T_0(x_{t-1}) = x_{t-1}, \tag{218}$$

$$A_0(\eta_0(x_t)) = 0. \tag{219}$$

Then, we define

$$\tilde{\eta}(x_t, x_0) := \eta_0(x_t) + \frac{\sqrt{\bar{\alpha}_{t-1}}}{1 - \bar{\alpha}_{t-1}} x_0 = \frac{\sqrt{\bar{\alpha}_{t-1}}}{1 - \bar{\alpha}_{t-1}} x_0, \tag{220}$$

so that

$$\eta(x_t, x_0)^T T(x_{t-1}) = \tilde{\eta}(x_t, x_0)^T x_{t-1}. \tag{221}$$

For the Beta distribution, we use the same log-partition form as above. Hence,

$$A(\eta(x_t, x_0)) = A(\tilde{\eta}(x_t, x_0)) \tag{222}$$

$$= \log \int_{\mathbb{R}^d} \exp\left(-\frac{\|x_{t-1}\|^2}{2(1-\bar{\alpha}_{t-1})}\right) \exp\left(-\frac{1-\beta_t}{2\beta_t}\|x_{t-1}\|^2\right) \exp\big(\tilde{\eta}(x_t, x_0)^T x_{t-1}\big)\, dx_{t-1}. \tag{223}$$

Equivalently,

$$A(\tilde{\eta}(x_t, x_0)) = \log \int_{\mathbb{R}^d} \exp\left(-\frac{1}{2}\left(\frac{1}{1-\bar{\alpha}_{t-1}} + \frac{1-\beta_t}{\beta_t}\right)\|x_{t-1}\|^2 + \tilde{\eta}(x_t, x_0)^T x_{t-1}\right) dx_{t-1}. \tag{224}$$

Define

$$\sigma_t^2 := \frac{1}{\frac{1}{1-\bar{\alpha}_{t-1}} + \frac{1-\beta_t}{\beta_t}} = \frac{\beta_t(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}. \tag{225}$$

Using the Gaussian integral identity, we obtain

$$A(\tilde{\eta}(x_t, x_0)) = \log\left((\sqrt{2\pi\sigma_t^2})^d\right) + \frac{\sigma_t^2}{2}\|\tilde{\eta}(x_t, x_0)\|^2. \tag{226}$$

Therefore,

$$D_{KL}(\tilde{q}(x_{t-1} \mid x_t, x_0) \,\|\, p_\theta(x_{t-1} \mid x_t)) \tag{227}$$

$$= A(\eta_\theta(x_t, t)) - A(\tilde{\eta}(x_t, x_0)) - (\eta_\theta(x_t, t) - \tilde{\eta}(x_t, x_0))^T \nabla A(\tilde{\eta}(x_t, x_0)) \tag{228}$$

$$= \frac{\sigma_t^2}{2}\left\|\eta_\theta(x_t, t) - \tilde{\eta}(x_t, x_0)\right\|^2. \tag{229}$$

Next, we express $\tilde{\eta}(x_t, x_0)$ in terms of the latent variable K_t. From the forward process,

$$x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1-\bar{\alpha}_t}\, K_t, \tag{230}$$

which implies

$$x_0 = \frac{x_t - \sqrt{1-\bar{\alpha}_t}\, K_t}{\sqrt{\bar{\alpha}_t}}. \tag{231}$$

Substituting this into $\tilde{\eta}(x_t, x_0)$ gives

$$\tilde{\eta}(x_t, x_0) = \frac{\sqrt{\bar{\alpha}_{t-1}}}{1-\bar{\alpha}_{t-1}} \cdot \frac{x_t - \sqrt{1-\bar{\alpha}_t}\, K_t}{\sqrt{\bar{\alpha}_t}} \tag{232}$$

$$= \frac{1}{(1-\bar{\alpha}_{t-1})\sqrt{\alpha_t}}\, x_t - \frac{\sqrt{1-\bar{\alpha}_t}}{(1-\bar{\alpha}_{t-1})\sqrt{\alpha_t}}\, K_t. \tag{233}$$

Therefore, we can write

$$\tilde{\eta}(x_t, x_0) = \tilde{\eta}_x(x_t) - b_t K_t, \tag{234}$$

where

$$\tilde{\eta}_x(x_t) = \frac{1}{(1-\bar{\alpha}_{t-1})\sqrt{\alpha_t}}\, x_t, \tag{235}$$

$$b_t = \frac{\sqrt{1-\bar{\alpha}_t}}{(1-\bar{\alpha}_{t-1})\sqrt{\alpha_t}}. \tag{236}$$

We parameterize the model natural parameter in the same form:

$$\eta_\theta(x_t, t) = \tilde{\eta}_x(x_t) - b_t K_\theta(x_t, t), \tag{237}$$

Substituting $\tilde{\eta}(x_t, x_0) = \tilde{\eta}_x(x_t) - b_t K_t$ and $\eta_\theta(x_t, t) = \tilde{\eta}_x(x_t) - b_t K_\theta(x_t, t)$ into (229), we obtain

$$D_{KL}(\tilde{q}(x_{t-1} \mid x_t, x_0) \,\|\, p_\theta(x_{t-1} \mid x_t)) \tag{238}$$

$$= \frac{\sigma_t^2}{2} \left\| (\tilde{\eta}_x(x_t) - b_t K_\theta(x_t, t)) - (\tilde{\eta}_x(x_t) - b_t K_t) \right\|^2 \tag{239}$$

$$= \frac{\sigma_t^2 b_t^2}{2} \|K_\theta(x_t, t) - K_t\|^2. \tag{240}$$

Since

$$\sigma_t^2 b_t^2 = \frac{\beta_t(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} \cdot \frac{1 - \bar{\alpha}_t}{\alpha_t(1 - \bar{\alpha}_{t-1})^2} \tag{241}$$

$$= \frac{\beta_t}{\alpha_t(1 - \bar{\alpha}_{t-1})}, \tag{242}$$

it follows that

$$D_{KL}(\tilde{q}(x_{t-1} \mid x_t, x_0) \,\|\, p_\theta(x_{t-1} \mid x_t)) = \frac{\beta_t}{2\alpha_t(1 - \bar{\alpha}_{t-1})} \|K_\theta(x_t, t) - K_t\|^2. \tag{243}$$

Therefore, a natural training objective is

$$\tilde{L} = \mathbb{E}_q \Bigg[D_{KL}(q(x_T \mid x_0) \| p(x_T)) + \sum_{t>1} \frac{\beta_t}{2\alpha_t(1 - \bar{\alpha}_{t-1})} \|K_\theta(x_t, t) - K_t\|^2 - \log p_\theta(x_0 \mid x_1) \Bigg].$$

3.6 Chi-square Distribution

Let $\varepsilon_1, \varepsilon_2, \ldots, \varepsilon_T$ be a sequence of i.i.d. random vectors such that each coordinate is distributed as χ_ν^2, applied component-wise, with mean

$$\mu = \mathbb{E}[\chi_\nu^2] = \nu > 0.$$

For a fixed variance schedule $(\beta_t)_{t=1}^T$ with $\alpha_t = 1 - \beta_t$, we define the forward process

$$x_t = \sqrt{\alpha_t}\, x_{t-1} + \sqrt{\beta_t}\, (\varepsilon_t - \mu \mathbf{1}). \tag{244}$$

Then,

$$q(x_t \mid x_{t-1}) = \frac{1}{(\sqrt{\beta_t})^d} P_\varepsilon \left(\frac{x_t - \sqrt{\alpha_t} x_{t-1} + \sqrt{\beta_t} \mu \mathbf{1}}{\sqrt{\beta_t}} \right), \tag{245}$$

where

$$P_\varepsilon(z) \propto \prod_{i=1}^d z_i^{\nu/2-1} \exp(-z_i/2) \mathbf{1}_{\{z_i > 0\}}. \tag{246}$$

Hence,

$$q(x_t \mid x_{t-1}) \propto \frac{1}{(\sqrt{\beta_t})^d} \prod_{i=1}^d z_{t,i}^{\nu/2-1} \exp(-z_{t,i}/2)\, \mathbf{1}_{\{z_{t,i} > 0\}}, \tag{247}$$

where

$$z_t = \frac{x_t - \sqrt{\alpha_t} x_{t-1} + \sqrt{\beta_t} \mu \mathbf{1}}{\sqrt{\beta_t}} = a_t - b_t x_{t-1}, \tag{248}$$

with

$$a_t = \frac{x_t + \sqrt{\beta_t} \mu \mathbf{1}}{\sqrt{\beta_t}}, \qquad b_t = \sqrt{\frac{\alpha_t}{\beta_t}}. \tag{249}$$

Therefore,

$$q(x_t \mid x_{t-1}) = \frac{1}{(\sqrt{\beta_t})^d} \prod_{i=1}^{d} (a_{t,i} - b_t x_{t-1,i})^{\nu/2-1} \exp\Big(-\tfrac{1}{2}(a_{t,i} - b_t x_{t-1,i})\Big) \mathbf{1}_{\{a_{t,i} - b_t x_{t-1,i} > 0\}} \tag{250}$$

$$= \frac{1}{(\sqrt{\beta_t})^d} \exp\Big(-\tfrac{1}{2}\mathbf{1}^T a_t\Big) \prod_{i=1}^{d} (a_{t,i} - b_t x_{t-1,i})^{\nu/2-1} \mathbf{1}_{\{a_{t,i} - b_t x_{t-1,i} > 0\}} \exp\Big(\tfrac{1}{2} b_t \mathbf{1}^T x_{t-1}\Big). \tag{251}$$

Thus, we write

$$q(x_t \mid x_{t-1}) = g_{\chi^2}(x_t, x_{t-1}) \exp\big(\eta_0(x_t)^T T_0(x_{t-1})\big), \tag{252}$$

where

$$g_{\chi^2}(x_t, x_{t-1}) = \frac{1}{(\sqrt{\beta_t})^d} \exp\Big(-\tfrac{1}{2}\mathbf{1}^T a_t\Big) \prod_{i=1}^{d} (a_{t,i} - b_t x_{t-1,i})^{\nu/2-1} \mathbf{1}_{\{a_{t,i} - b_t x_{t-1,i} > 0\}}, \tag{253}$$

$$\eta_0(x_t) = \frac{b_t}{2}\mathbf{1} = \frac{1}{2}\sqrt{\frac{\alpha_t}{\beta_t}}\,\mathbf{1}, \tag{254}$$

$$T_0(x_{t-1}) = x_{t-1}, \tag{255}$$

$$A_0(\eta_0(x_t)) = 0. \tag{256}$$

Then, we define

$$\tilde{\eta}(x_t, x_0) := \eta_0(x_t) + \frac{\sqrt{\bar{\alpha}_{t-1}}}{1 - \bar{\alpha}_{t-1}} x_0 \tag{257}$$

$$= \frac{1}{2}\sqrt{\frac{\alpha_t}{\beta_t}}\,\mathbf{1} + \frac{\sqrt{\bar{\alpha}_{t-1}}}{1 - \bar{\alpha}_{t-1}} x_0, \tag{258}$$

so that

$$\eta(x_t, x_0)^T T(x_{t-1}) = \tilde{\eta}(x_t, x_0)^T x_{t-1}. \tag{259}$$

For the chi-square distribution, we use the same log-partition form as above. Hence,

$$A(\eta(x_t, x_0)) = A(\tilde{\eta}(x_t, x_0)) \tag{260}$$

$$= \log \int_{\mathbb{R}^d} \exp\left(-\frac{\|x_{t-1}\|^2}{2(1 - \bar{\alpha}_{t-1})} \right) \exp\left(-\frac{1 - \beta_t}{2\beta_t} \|x_{t-1}\|^2 \right) \exp\big(\tilde{\eta}(x_t, x_0)^T x_{t-1}\big)\, dx_{t-1}. \tag{261}$$

Equivalently,

$$A(\tilde{\eta}(x_t, x_0)) = \log \int_{\mathbb{R}^d} \exp\left(-\frac{1}{2}\left(\frac{1}{1 - \bar{\alpha}_{t-1}} + \frac{1 - \beta_t}{\beta_t} \right) \|x_{t-1}\|^2 + \tilde{\eta}(x_t, x_0)^T x_{t-1} \right) dx_{t-1}. \tag{262}$$

Define

$$\sigma_t^2 := \frac{1}{\frac{1}{1 - \bar{\alpha}_{t-1}} + \frac{1 - \beta_t}{\beta_t}} = \frac{\beta_t (1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t}. \tag{263}$$

Using the Gaussian integral identity, we obtain

$$A(\tilde{\eta}(x_t, x_0)) = \log\left((\sqrt{2\pi\sigma_t^2})^d \right) + \frac{\sigma_t^2}{2} \|\tilde{\eta}(x_t, x_0)\|^2. \tag{264}$$

Therefore,

$$D_{KL}(\tilde{q}(x_{t-1} \mid x_t, x_0) \,\|\, p_\theta(x_{t-1} \mid x_t)) \tag{265}$$

$$= A(\eta_\theta(x_t, t)) - A(\tilde{\eta}(x_t, x_0)) - (\eta_\theta(x_t, t) - \tilde{\eta}(x_t, x_0))^T \nabla A(\tilde{\eta}(x_t, x_0)) \tag{266}$$

$$= \frac{\sigma_t^2}{2} \left\| \eta_\theta(x_t, t) - \tilde{\eta}(x_t, x_0) \right\|^2. \tag{267}$$

Next, we express $\tilde{\eta}(x_t, x_0)$ in terms of the latent variable K_t. From the forward process,

$$x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1-\bar{\alpha}_t}\, K_t, \tag{268}$$

which implies

$$x_0 = \frac{x_t - \sqrt{1-\bar{\alpha}_t}\, K_t}{\sqrt{\bar{\alpha}_t}}. \tag{269}$$

Substituting this into $\tilde{\eta}(x_t, x_0)$ gives

$$\tilde{\eta}(x_t, x_0) = \frac{1}{2}\sqrt{\frac{\alpha_t}{\beta_t}}\,\mathbf{1} + \frac{\sqrt{\bar{\alpha}_{t-1}}}{1-\bar{\alpha}_{t-1}} \cdot \frac{x_t - \sqrt{1-\bar{\alpha}_t}\, K_t}{\sqrt{\bar{\alpha}_t}} \tag{270}$$

$$= \frac{1}{2}\sqrt{\frac{\alpha_t}{\beta_t}}\,\mathbf{1} + \frac{1}{(1-\bar{\alpha}_{t-1})\sqrt{\alpha_t}}\, x_t - \frac{\sqrt{1-\bar{\alpha}_t}}{(1-\bar{\alpha}_{t-1})\sqrt{\alpha_t}}\, K_t. \tag{271}$$

Therefore, we can write

$$\tilde{\eta}(x_t, x_0) = \tilde{\eta}_x(x_t) - b_t K_t, \tag{272}$$

where

$$\tilde{\eta}_x(x_t) = \frac{1}{2}\sqrt{\frac{\alpha_t}{\beta_t}}\,\mathbf{1} + \frac{1}{(1-\bar{\alpha}_{t-1})\sqrt{\alpha_t}}\, x_t, \tag{273}$$

$$b_t = \frac{\sqrt{1-\bar{\alpha}_t}}{(1-\bar{\alpha}_{t-1})\sqrt{\alpha_t}}. \tag{274}$$

We parameterize the model natural parameter in the same form:

$$\eta_\theta(x_t, t) = \tilde{\eta}_x(x_t) - b_t K_\theta(x_t, t), \tag{275}$$

Substituting $\tilde{\eta}(x_t, x_0) = \tilde{\eta}_x(x_t) - b_t K_t$ and $\eta_\theta(x_t, t) = \tilde{\eta}_x(x_t) - b_t K_\theta(x_t, t)$ into (267), we obtain

$$D_{KL}(\tilde{q}(x_{t-1} \mid x_t, x_0) \,\|\, p_\theta(x_{t-1} \mid x_t)) \tag{276}$$

$$= \frac{\sigma_t^2}{2}\, \|(\tilde{\eta}_x(x_t) - b_t K_\theta(x_t, t)) - (\tilde{\eta}_x(x_t) - b_t K_t)\|^2 \tag{277}$$

$$= \frac{\sigma_t^2 b_t^2}{2} \|K_\theta(x_t, t) - K_t\|^2. \tag{278}$$

Since

$$\sigma_t^2 b_t^2 = \frac{\beta_t(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t} \cdot \frac{1-\bar{\alpha}_t}{\alpha_t(1-\bar{\alpha}_{t-1})^2} \tag{279}$$

$$= \frac{\beta_t}{\alpha_t(1-\bar{\alpha}_{t-1})}, \tag{280}$$

it follows that

$$D_{KL}(\tilde{q}(x_{t-1} \mid x_t, x_0) \,\|\, p_\theta(x_{t-1} \mid x_t)) = \frac{\beta_t}{2\alpha_t(1-\bar{\alpha}_{t-1})} \|K_\theta(x_t, t) - K_t\|^2. \tag{281}$$

Therefore, a natural training objective is

$$\tilde{L} = \mathbb{E}_q\Bigg[D_{KL}(q(x_T \mid x_0)\|p(x_T)) + \sum_{t>1} \frac{\beta_t}{2\alpha_t(1-\bar{\alpha}_{t-1})} \|K_\theta(x_t, t) - K_t\|^2 - \log p_\theta(x_0 \mid x_1)\Bigg]. \tag{282}$$

3.7 Dirichlet Distribution

Let $\varepsilon_1, \varepsilon_2, \ldots, \varepsilon_T$ be a sequence of i.i.d. random vectors distributed as $\mathrm{Dir}(\boldsymbol{\gamma})$ on the simplex

$$\Delta^{d-1} = \Big\{ z \in \mathbb{R}^d : \; z_i > 0, \; \sum_{i=1}^{d} z_i = 1 \Big\}.$$

Their mean is

$$\boldsymbol{\mu} = \mathbb{E}[\varepsilon_t] = \frac{\boldsymbol{\gamma}}{\gamma_0}, \qquad \gamma_0 = \sum_{i=1}^{d} \gamma_i.$$

For a fixed variance schedule $(\beta_t)_{t=1}^T$ with $\alpha_t = 1 - \beta_t$, we define the forward process

$$x_t = \sqrt{\alpha_t}\, x_{t-1} + \sqrt{\beta_t}\, (\varepsilon_t - \boldsymbol{\mu}). \tag{283}$$

Then,

$$q(x_t \mid x_{t-1}) = \frac{1}{(\sqrt{\beta_t})^d} P_\varepsilon \left(\frac{x_t - \sqrt{\alpha_t} x_{t-1} + \sqrt{\beta_t}\boldsymbol{\mu}}{\sqrt{\beta_t}} \right), \tag{284}$$

where the Dirichlet density satisfies

$$P_\varepsilon(z) \propto \prod_{i=1}^{d} z_i^{\gamma_i - 1}\, \mathbf{1}_{\{z \in \Delta^{d-1}\}}. \tag{285}$$

Hence,

$$q(x_t \mid x_{t-1}) \propto \frac{1}{(\sqrt{\beta_t})^d} \prod_{i=1}^{d} z_{t,i}^{\gamma_i - 1}\, \mathbf{1}_{\{z_t \in \Delta^{d-1}\}}, \tag{286}$$

where

$$z_t = \frac{x_t - \sqrt{\alpha_t} x_{t-1} + \sqrt{\beta_t}\boldsymbol{\mu}}{\sqrt{\beta_t}} = a_t - b_t x_{t-1}, \tag{287}$$

with

$$a_t = \frac{x_t + \sqrt{\beta_t}\boldsymbol{\mu}}{\sqrt{\beta_t}}, \qquad b_t = \sqrt{\frac{\alpha_t}{\beta_t}}. \tag{288}$$

Therefore,

$$q(x_t \mid x_{t-1}) = \frac{1}{(\sqrt{\beta_t})^d} \prod_{i=1}^{d} (a_{t,i} - b_t x_{t-1,i})^{\gamma_i - 1}\, \mathbf{1}_{\{a_t - b_t x_{t-1} \in \Delta^{d-1}\}}. \tag{289}$$

Thus, we write

$$q(x_t \mid x_{t-1}) = g_D(x_t, x_{t-1}) \exp\big(\eta_0(x_t)^T T_0(x_{t-1})\big), \tag{290}$$

where

$$g_D(x_t, x_{t-1}) = \frac{1}{(\sqrt{\beta_t})^d} \prod_{i=1}^{d} (a_{t,i} - b_t x_{t-1,i})^{\gamma_i - 1}\, \mathbf{1}_{\{a_t - b_t x_{t-1} \in \Delta^{d-1}\}}, \tag{291}$$

$$\eta_0(x_t) = 0, \tag{292}$$

$$T_0(x_{t-1}) = x_{t-1}, \tag{293}$$

$$A_0(\eta_0(x_t)) = 0. \tag{294}$$

Then, we define

$$\tilde{\eta}(x_t, x_0) := \eta_0(x_t) + \frac{\sqrt{\bar{\alpha}_{t-1}}}{1 - \bar{\alpha}_{t-1}} x_0 \tag{295}$$

$$= \frac{\sqrt{\bar{\alpha}_{t-1}}}{1 - \bar{\alpha}_{t-1}} x_0, \tag{296}$$

so that

$$\eta(x_t, x_0)^T T(x_{t-1}) = \tilde{\eta}(x_t, x_0)^T x_{t-1}. \tag{297}$$

For the distribution D, we use the same log-partition form as above. Hence,

$$A(\eta(x_t, x_0)) = A(\tilde{\eta}(x_t, x_0)) \tag{298}$$

$$= \log \int_{\mathbb{R}^d} \exp\left(-\frac{\|x_{t-1}\|^2}{2(1-\bar{\alpha}_{t-1})}\right) \exp\left(-\frac{1-\beta_t}{2\beta_t}\|x_{t-1}\|^2\right) \exp\big(\tilde{\eta}(x_t, x_0)^T x_{t-1}\big)\, dx_{t-1}. \tag{299}$$

Equivalently,

$$A(\tilde{\eta}(x_t, x_0)) = \log \int_{\mathbb{R}^d} \exp\left(-\frac{1}{2}\left(\frac{1}{1-\bar{\alpha}_{t-1}} + \frac{1-\beta_t}{\beta_t}\right)\|x_{t-1}\|^2 + \tilde{\eta}(x_t, x_0)^T x_{t-1}\right) dx_{t-1}. \tag{300}$$

Define

$$\sigma_t^2 := \frac{1}{\frac{1}{1-\bar{\alpha}_{t-1}} + \frac{1-\beta_t}{\beta_t}} = \frac{\beta_t(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}. \tag{301}$$

Using the Gaussian integral identity, we obtain

$$A(\tilde{\eta}(x_t, x_0)) = \log\left((\sqrt{2\pi\sigma_t^2})^d\right) + \frac{\sigma_t^2}{2}\|\tilde{\eta}(x_t, x_0)\|^2. \tag{302}$$

Therefore,

$$D_{KL}(\tilde{q}(x_{t-1} \mid x_t, x_0) \,\|\, p_\theta(x_{t-1} \mid x_t)) \tag{303}$$

$$= A(\eta_\theta(x_t, t)) - A(\tilde{\eta}(x_t, x_0)) - (\eta_\theta(x_t, t) - \tilde{\eta}(x_t, x_0))^T \nabla A(\tilde{\eta}(x_t, x_0)) \tag{304}$$

$$= \frac{\sigma_t^2}{2}\left\|\eta_\theta(x_t, t) - \tilde{\eta}(x_t, x_0)\right\|^2. \tag{305}$$

Next, we express $\tilde{\eta}(x_t, x_0)$ in terms of the latent variable K_t. From the forward process,

$$x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1-\bar{\alpha}_t}\, K_t, \tag{306}$$

which implies

$$x_0 = \frac{x_t - \sqrt{1-\bar{\alpha}_t}\, K_t}{\sqrt{\bar{\alpha}_t}}. \tag{307}$$

Substituting this into $\tilde{\eta}(x_t, x_0)$ gives

$$\tilde{\eta}(x_t, x_0) = \frac{\sqrt{\bar{\alpha}_{t-1}}}{1-\bar{\alpha}_{t-1}} \cdot \frac{x_t - \sqrt{1-\bar{\alpha}_t}\, K_t}{\sqrt{\bar{\alpha}_t}} \tag{308}$$

$$= \frac{1}{(1-\bar{\alpha}_{t-1})\sqrt{\alpha_t}}\, x_t - \frac{\sqrt{1-\bar{\alpha}_t}}{(1-\bar{\alpha}_{t-1})\sqrt{\alpha_t}}\, K_t. \tag{309}$$

Therefore, we can write

$$\tilde{\eta}(x_t, x_0) = \tilde{\eta}_x(x_t) - b_t K_t, \tag{310}$$

where

$$\tilde{\eta}_x(x_t) = \frac{1}{(1-\bar{\alpha}_{t-1})\sqrt{\alpha_t}}\, x_t, \tag{311}$$

$$b_t = \frac{\sqrt{1-\bar{\alpha}_t}}{(1-\bar{\alpha}_{t-1})\sqrt{\alpha_t}}. \tag{312}$$

We parameterize the model natural parameter in the same form:

$$\eta_\theta(x_t, t) = \tilde{\eta}_x(x_t) - b_t K_\theta(x_t, t), \tag{313}$$

Substituting $\tilde{\eta}(x_t, x_0) = \tilde{\eta}_x(x_t) - b_t K_t$ and $\eta_\theta(x_t, t) = \tilde{\eta}_x(x_t) - b_t K_\theta(x_t, t)$ into (305), we obtain

$$D_{KL}(\tilde{q}(x_{t-1} \mid x_t, x_0) \,\|\, p_\theta(x_{t-1} \mid x_t)) \tag{314}$$

$$= \frac{\sigma_t^2}{2} \left\|(\tilde{\eta}_x(x_t) - b_t K_\theta(x_t, t)) - (\tilde{\eta}_x(x_t) - b_t K_t)\right\|^2 \tag{315}$$

$$= \frac{\sigma_t^2 b_t^2}{2} \|K_\theta(x_t, t) - K_t\|^2. \tag{316}$$

Since

$$\sigma_t^2 b_t^2 = \frac{\beta_t(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} \cdot \frac{1 - \bar{\alpha}_t}{\alpha_t(1 - \bar{\alpha}_{t-1})^2} \tag{317}$$

$$= \frac{\beta_t}{\alpha_t(1 - \bar{\alpha}_{t-1})}, \tag{318}$$

it follows that

$$D_{KL}(\tilde{q}(x_{t-1} \mid x_t, x_0) \,\|\, p_\theta(x_{t-1} \mid x_t)) = \frac{\beta_t}{2\alpha_t(1 - \bar{\alpha}_{t-1})} \|K_\theta(x_t, t) - K_t\|^2. \tag{319}$$

Therefore, a natural training objective is

$$\tilde{L} = \mathbb{E}_q\Bigg[D_{KL}(q(x_T \mid x_0)\|p(x_T)) + \sum_{t>1} \frac{\beta_t}{2\alpha_t(1 - \bar{\alpha}_{t-1})} \|K_\theta(x_t, t) - K_t\|^2 - \log p_\theta(x_0 \mid x_1)\Bigg]. \tag{320}$$

3.8 Inverse-Gamma Distribution

Let $\varepsilon_1, \varepsilon_2, \ldots, \varepsilon_T$ be a sequence of i.i.d. random vectors such that each coordinate follows $\text{Inv-Gamma}(\kappa, \theta)$, applied component-wise, with mean

$$\mu = \mathbb{E}[\text{Inv-Gamma}(\kappa, \theta)] = \frac{\theta}{\kappa - 1}, \qquad \kappa > 1.$$

For a fixed variance schedule $(\beta_t)_{t=1}^T$ with $\alpha_t = 1 - \beta_t$, we define the forward process

$$x_t = \sqrt{\alpha_t}\, x_{t-1} + \sqrt{\beta_t}\, (\varepsilon_t - \mu \mathbf{1}). \tag{321}$$

Then,

$$q(x_t \mid x_{t-1}) = \frac{1}{(\sqrt{\beta_t})^d} P_\varepsilon\left(\frac{x_t - \sqrt{\alpha_t} x_{t-1} + \sqrt{\beta_t}\mu \mathbf{1}}{\sqrt{\beta_t}}\right), \tag{322}$$

where the inverse-Gamma density satisfies

$$P_\varepsilon(z) \propto \prod_{i=1}^d z_i^{-(\kappa+1)} \exp\Big(-\frac{\theta}{z_i}\Big) \mathbf{1}_{\{z_i > 0\}}. \tag{323}$$

Hence,

$$q(x_t \mid x_{t-1}) \propto \frac{1}{(\sqrt{\beta_t})^d} \prod_{i=1}^d z_{t,i}^{-(\kappa+1)} \exp\Big(-\frac{\theta}{z_{t,i}}\Big) \mathbf{1}_{\{z_{t,i} > 0\}}, \tag{324}$$

where

$$z_t = \frac{x_t - \sqrt{\alpha_t} x_{t-1} + \sqrt{\beta_t}\mu \mathbf{1}}{\sqrt{\beta_t}} = a_t - b_t x_{t-1}, \tag{325}$$

with

$$a_t = \frac{x_t + \sqrt{\beta_t}\mu \mathbf{1}}{\sqrt{\beta_t}}, \qquad b_t = \sqrt{\frac{\alpha_t}{\beta_t}}. \tag{326}$$

Therefore,

$$q(x_t \mid x_{t-1}) = \frac{1}{(\sqrt{\beta_t})^d} \prod_{i=1}^{d} (a_{t,i} - b_t x_{t-1,i})^{-(\kappa+1)} \exp\Big(- \frac{\theta}{a_{t,i} - b_t x_{t-1,i}} \Big) \mathbf{1}_{\{a_{t,i} - b_t x_{t-1,i} > 0\}}. \tag{327}$$

Thus, we write

$$q(x_t \mid x_{t-1}) = g_{IG}(x_t, x_{t-1}) \exp\big(\eta_0(x_t)^T T_0(x_{t-1})\big), \tag{328}$$

where

$$g_{IG}(x_t, x_{t-1}) = \frac{1}{(\sqrt{\beta_t})^d} \prod_{i=1}^{d} (a_{t,i} - b_t x_{t-1,i})^{-(\kappa+1)} \exp\Big(- \frac{\theta}{a_{t,i} - b_t x_{t-1,i}} \Big) \mathbf{1}_{\{a_{t,i} - b_t x_{t-1,i} > 0\}}, \tag{329}$$

$$\eta_0(x_t) = 0, \tag{330}$$

$$T_0(x_{t-1}) = x_{t-1}, \tag{331}$$

$$A_0(\eta_0(x_t)) = 0. \tag{332}$$

Then, we define

$$\tilde{\eta}(x_t, x_0) := \eta_0(x_t) + \frac{\sqrt{\bar{\alpha}_{t-1}}}{1 - \bar{\alpha}_{t-1}} x_0 \tag{333}$$

$$= \frac{\sqrt{\bar{\alpha}_{t-1}}}{1 - \bar{\alpha}_{t-1}} x_0, \tag{334}$$

so that

$$\eta(x_t, x_0)^T T(x_{t-1}) = \tilde{\eta}(x_t, x_0)^T x_{t-1}. \tag{335}$$

For the inverse Gaussian distribution, we use the same log-partition form as above. Hence,

$$A(\eta(x_t, x_0)) = A(\tilde{\eta}(x_t, x_0)) \tag{336}$$

$$= \log \int_{\mathbb{R}^d} \exp\left(- \frac{\|x_{t-1}\|^2}{2(1 - \bar{\alpha}_{t-1})} \right) \exp\left(- \frac{1 - \beta_t}{2\beta_t} \|x_{t-1}\|^2 \right) \exp\big(\tilde{\eta}(x_t, x_0)^T x_{t-1}\big) \, dx_{t-1}. \tag{337}$$

Equivalently,

$$A(\tilde{\eta}(x_t, x_0)) = \log \int_{\mathbb{R}^d} \exp\left(- \frac{1}{2} \left(\frac{1}{1 - \bar{\alpha}_{t-1}} + \frac{1 - \beta_t}{\beta_t} \right) \|x_{t-1}\|^2 + \tilde{\eta}(x_t, x_0)^T x_{t-1} \right) \, dx_{t-1}. \tag{338}$$

Define

$$\sigma_t^2 := \frac{1}{\frac{1}{1 - \bar{\alpha}_{t-1}} + \frac{1 - \beta_t}{\beta_t}} = \frac{\beta_t (1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t}. \tag{339}$$

Using the Gaussian integral identity, we obtain

$$A(\tilde{\eta}(x_t, x_0)) = \log \left((\sqrt{2\pi\sigma_t^2})^d \right) + \frac{\sigma_t^2}{2} \|\tilde{\eta}(x_t, x_0)\|^2. \tag{340}$$

Therefore,

$$D_{KL}(\tilde{q}(x_{t-1} \mid x_t, x_0) \,\|\, p_\theta(x_{t-1} \mid x_t)) \tag{341}$$

$$= A(\eta_\theta(x_t, t)) - A(\tilde{\eta}(x_t, x_0)) - (\eta_\theta(x_t, t) - \tilde{\eta}(x_t, x_0))^T \nabla A(\tilde{\eta}(x_t, x_0)) \tag{342}$$

$$= \frac{\sigma_t^2}{2} \left\| \eta_\theta(x_t, t) - \tilde{\eta}(x_t, x_0) \right\|^2. \tag{343}$$

Next, we express $\tilde{\eta}(x_t, x_0)$ in terms of the latent variable K_t. From the forward process,

$$x_t = \sqrt{\bar{\alpha}_t} \, x_0 + \sqrt{1 - \bar{\alpha}_t} \, K_t, \tag{344}$$

which implies

$$x_0 = \frac{x_t - \sqrt{1-\bar{\alpha}_t}\, K_t}{\sqrt{\bar{\alpha}_t}}. \tag{345}$$

Substituting this into $\tilde{\eta}(x_t, x_0)$ gives

$$\tilde{\eta}(x_t, x_0) = \frac{\sqrt{\bar{\alpha}_{t-1}}}{1-\bar{\alpha}_{t-1}} \cdot \frac{x_t - \sqrt{1-\bar{\alpha}_t}\, K_t}{\sqrt{\bar{\alpha}_t}} \tag{346}$$

$$= \frac{1}{(1-\bar{\alpha}_{t-1})\sqrt{\alpha_t}}\, x_t - \frac{\sqrt{1-\bar{\alpha}_t}}{(1-\bar{\alpha}_{t-1})\sqrt{\alpha_t}}\, K_t. \tag{347}$$

Therefore, we can write

$$\tilde{\eta}(x_t, x_0) = \tilde{\eta}_x(x_t) - b_t K_t, \tag{348}$$

where

$$\tilde{\eta}_x(x_t) = \frac{1}{(1-\bar{\alpha}_{t-1})\sqrt{\alpha_t}}\, x_t, \tag{349}$$

$$b_t = \frac{\sqrt{1-\bar{\alpha}_t}}{(1-\bar{\alpha}_{t-1})\sqrt{\alpha_t}}. \tag{350}$$

We parameterize the model natural parameter in the same form:

$$\eta_\theta(x_t, t) = \tilde{\eta}_x(x_t) - b_t K_\theta(x_t, t), \tag{351}$$

Substituting $\tilde{\eta}(x_t, x_0) = \tilde{\eta}_x(x_t) - b_t K_t$ and $\eta_\theta(x_t, t) = \tilde{\eta}_x(x_t) - b_t K_\theta(x_t, t)$ into (343), we obtain

$$D_{KL}(\tilde{q}(x_{t-1} \mid x_t, x_0) \,\|\, p_\theta(x_{t-1} \mid x_t)) \tag{352}$$

$$= \frac{\sigma_t^2}{2} \left\| (\tilde{\eta}_x(x_t) - b_t K_\theta(x_t, t)) - (\tilde{\eta}_x(x_t) - b_t K_t) \right\|^2 \tag{353}$$

$$= \frac{\sigma_t^2 b_t^2}{2} \|K_\theta(x_t, t) - K_t\|^2. \tag{354}$$

Since

$$\sigma_t^2 b_t^2 = \frac{\beta_t(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t} \cdot \frac{1-\bar{\alpha}_t}{\alpha_t(1-\bar{\alpha}_{t-1})^2} \tag{355}$$

$$= \frac{\beta_t}{\alpha_t(1-\bar{\alpha}_{t-1})}, \tag{356}$$

it follows that

$$D_{KL}(\tilde{q}(x_{t-1} \mid x_t, x_0) \,\|\, p_\theta(x_{t-1} \mid x_t)) = \frac{\beta_t}{2\alpha_t(1-\bar{\alpha}_{t-1})} \|K_\theta(x_t, t) - K_t\|^2. \tag{357}$$

Therefore, a natural training objective is

$$\tilde{L} = \mathbb{E}_q \Bigg[D_{KL}(q(x_T \mid x_0) \| p(x_T)) + \sum_{t>1} \frac{\beta_t}{2\alpha_t(1-\bar{\alpha}_{t-1})} \|K_\theta(x_t, t) - K_t\|^2 - \log p_\theta(x_0 \mid x_1) \Bigg]. \tag{358}$$

3.9 Wishart Distribution

For the Wishart distribution $\mathcal{W}_p(\nu, \Sigma)$ on the cone of symmetric positive definite matrices $\mathbb{S}_+^p$, the density is

$$p_{\mathcal{W}}(Z) \propto |Z|^{(\nu-p-1)/2} \exp\Big(-\tfrac{1}{2}\mathrm{tr}(\Sigma^{-1} Z) \Big) \mathbf{1}_{\{Z \in \mathbb{S}_+^p\}}.$$

Let $\varepsilon_1, \ldots, \varepsilon_T$ be i.i.d. with $\varepsilon_t \sim \mathcal{W}_p(\nu, \Sigma)$ and mean

$$M := \mathbb{E}[\varepsilon_t] = \nu\Sigma.$$

For a fixed variance schedule $(\beta_t)_{t=1}^T$ with $\alpha_t = 1 - \beta_t$, we define the forward process on matrices

$$X_t = \sqrt{\alpha_t}\, X_{t-1} + \sqrt{\beta_t}\,(\varepsilon_t - M). \tag{359}$$

Then the conditional density takes the form

$$q(X_t \mid X_{t-1}) = \frac{1}{(\sqrt{\beta_t})^m} P_\varepsilon\left(\frac{X_t - \sqrt{\alpha_t}X_{t-1} + \sqrt{\beta_t}M}{\sqrt{\beta_t}}\right), \tag{360}$$

where $m := p(p+1)/2$ is the dimension of the symmetric matrix space. Ignoring the global normalizing constant depending only on ν, Σ, we may write

$$q(X_t \mid X_{t-1}) = \frac{1}{(\sqrt{\beta_t})^m} |Z_t|^{(\nu-p-1)/2} \exp\Big(-\tfrac{1}{2}\mathrm{tr}(\Sigma^{-1}Z_t)\Big)\mathbf{1}_{\{Z_t \in \mathbb{S}_+^p\}}, \tag{361}$$

where

$$Z_t = \frac{X_t - \sqrt{\alpha_t}X_{t-1} + \sqrt{\beta_t}M}{\sqrt{\beta_t}} = A_t - b_t X_{t-1}, \tag{362}$$

with

$$A_t = \frac{X_t + \sqrt{\beta_t}M}{\sqrt{\beta_t}}, \qquad b_t = \sqrt{\frac{\alpha_t}{\beta_t}}. \tag{363}$$

Therefore,

$$q(X_t \mid X_{t-1}) = \frac{1}{(\sqrt{\beta_t})^m} |A_t - b_t X_{t-1}|^{(\nu-p-1)/2} \exp\Big(-\tfrac{1}{2}\mathrm{tr}(\Sigma^{-1}(A_t - b_t X_{t-1}))\Big)\mathbf{1}_{\{A_t - b_t X_{t-1} \in \mathbb{S}_+^p\}} \tag{364}$$

$$= \frac{1}{(\sqrt{\beta_t})^m} \exp\Big(-\tfrac{1}{2}\mathrm{tr}(\Sigma^{-1}A_t)\Big)|A_t - b_t X_{t-1}|^{(\nu-p-1)/2}\mathbf{1}_{\{A_t - b_t X_{t-1} \in \mathbb{S}_+^p\}} \tag{365}$$

$$\times \exp\Big(\tfrac{b_t}{2}\mathrm{tr}(\Sigma^{-1}X_{t-1})\Big). \tag{366}$$

Thus, we write

$$q(X_t \mid X_{t-1}) = g_W(X_t, X_{t-1}) \exp\big(\langle \eta_0(X_t), T_0(X_{t-1})\rangle\big), \tag{367}$$

where

$$g_W(X_t, X_{t-1}) = \frac{1}{(\sqrt{\beta_t})^m} \exp\Big(-\tfrac{1}{2}\mathrm{tr}(\Sigma^{-1}A_t)\Big)|A_t - b_t X_{t-1}|^{(\nu-p-1)/2}\mathbf{1}_{\{A_t - b_t X_{t-1} \in \mathbb{S}_+^p\}}, \tag{368}$$

$$\eta_0(X_t) = \frac{b_t}{2}\Sigma^{-1} = \frac{1}{2}\sqrt{\frac{\alpha_t}{\beta_t}}\,\Sigma^{-1}, \tag{369}$$

$$T_0(X_{t-1}) = X_{t-1}, \tag{370}$$

$$A_0(\eta_0(X_t)) = 0. \tag{371}$$

Then, we define

$$\tilde{\eta}(X_t, X_0) := \eta_0(X_t) + \frac{\sqrt{\bar{\alpha}_{t-1}}}{1 - \bar{\alpha}_{t-1}}\, X_0 \tag{372}$$

$$= \frac{1}{2}\sqrt{\frac{\alpha_t}{\beta_t}}\,\Sigma^{-1} + \frac{\sqrt{\bar{\alpha}_{t-1}}}{1 - \bar{\alpha}_{t-1}}\, X_0, \tag{373}$$

so that

$$\eta(X_t, X_0)^T T(X_{t-1}) = \big\langle \tilde{\eta}(X_t, X_0), X_{t-1}\big\rangle. \tag{374}$$

For the Wishart distribution, we use the same log-partition form as above. Hence,

$$A(\eta(X_t, X_0)) = A(\tilde{\eta}(X_t, X_0)) \tag{375}$$

$$= \log \int_{\mathbb{S}^p} \exp\left(-\frac{\|X_{t-1}\|_F^2}{2(1 - \bar{\alpha}_{t-1})}\right) \exp\left(-\frac{1 - \beta_t}{2\beta_t}\|X_{t-1}\|_F^2\right) \exp(\langle \tilde{\eta}(X_t, X_0), X_{t-1}\rangle)\, dX_{t-1}. \tag{376}$$

Equivalently,

$$A(\tilde{\eta}(X_t, X_0)) = \log \int_{\mathbb{S}^p} \exp\left(-\frac{1}{2}\left(\frac{1}{1-\bar{\alpha}_{t-1}} + \frac{1-\beta_t}{\beta_t}\right)\|X_{t-1}\|_F^2 + \langle \tilde{\eta}(X_t, X_0), X_{t-1}\rangle\right) dX_{t-1}. \tag{377}$$

Define

$$\sigma_t^2 := \frac{1}{\frac{1}{1-\bar{\alpha}_{t-1}} + \frac{1-\beta_t}{\beta_t}} = \frac{\beta_t(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}. \tag{378}$$

Using the Gaussian integral identity on the symmetric-matrix space, we obtain

$$A(\tilde{\eta}(X_t, X_0)) = \log\left((\sqrt{2\pi\sigma_t^2})^m\right) + \frac{\sigma_t^2}{2}\|\tilde{\eta}(X_t, X_0)\|_F^2, \tag{379}$$

where $m = \frac{p(p+1)}{2}$ is the dimension of $\mathbb{S}^p$.

Therefore,

$$\begin{aligned} &D_{KL}(\tilde{q}(X_{t-1} \mid X_t, X_0) \,\|\, p_\theta(X_{t-1} \mid X_t)) && (380)\\ &\quad = A(\eta_\theta(X_t, t)) - A(\tilde{\eta}(X_t, X_0)) - \left\langle \eta_\theta(X_t, t) - \tilde{\eta}(X_t, X_0), \nabla A(\tilde{\eta}(X_t, X_0))\right\rangle && (381)\\ &\quad = \frac{\sigma_t^2}{2}\left\|\eta_\theta(X_t, t) - \tilde{\eta}(X_t, X_0)\right\|_F^2. && (382) \end{aligned}$$

Next, we express $\tilde{\eta}(X_t, X_0)$ in terms of the latent variable K_t. From the forward process,

$$X_t = \sqrt{\bar{\alpha}_t}\, X_0 + \sqrt{1-\bar{\alpha}_t}\, K_t, \tag{383}$$

which implies

$$X_0 = \frac{X_t - \sqrt{1-\bar{\alpha}_t}\, K_t}{\sqrt{\bar{\alpha}_t}}. \tag{384}$$

Substituting this into $\tilde{\eta}(X_t, X_0)$ gives

$$\begin{aligned} \tilde{\eta}(X_t, X_0) &= \frac{1}{2}\sqrt{\frac{\alpha_t}{\beta_t}}\,\Sigma^{-1} + \frac{\sqrt{\bar{\alpha}_{t-1}}}{1-\bar{\alpha}_{t-1}} \cdot \frac{X_t - \sqrt{1-\bar{\alpha}_t}\, K_t}{\sqrt{\bar{\alpha}_t}} && (385)\\ &= \frac{1}{2}\sqrt{\frac{\alpha_t}{\beta_t}}\,\Sigma^{-1} + \frac{1}{(1-\bar{\alpha}_{t-1})\sqrt{\alpha_t}}\, X_t - \frac{\sqrt{1-\bar{\alpha}_t}}{(1-\bar{\alpha}_{t-1})\sqrt{\alpha_t}}\, K_t. && (386) \end{aligned}$$

Therefore, we can write

$$\tilde{\eta}(X_t, X_0) = \tilde{\eta}_X(X_t) - b_t^\eta K_t, \tag{387}$$

where

$$\begin{aligned} \tilde{\eta}_X(X_t) &= \frac{1}{2}\sqrt{\frac{\alpha_t}{\beta_t}}\,\Sigma^{-1} + \frac{1}{(1-\bar{\alpha}_{t-1})\sqrt{\alpha_t}}\, X_t, && (388)\\ b_t^\eta &= \frac{\sqrt{1-\bar{\alpha}_t}}{(1-\bar{\alpha}_{t-1})\sqrt{\alpha_t}}. && (389) \end{aligned}$$

We parameterize the model natural parameter in the same form:

$$\eta_\theta(X_t, t) = \tilde{\eta}_X(X_t) - b_t^\eta K_\theta(X_t, t), \tag{390}$$

Substituting $\tilde{\eta}(X_t, X_0) = \tilde{\eta}_X(X_t) - b_t^\eta K_t$ and $\eta_\theta(X_t, t) = \tilde{\eta}_X(X_t) - b_t^\eta K_\theta(X_t, t)$ into (382), we obtain

$$\begin{aligned} &D_{KL}(\tilde{q}(X_{t-1} \mid X_t, X_0) \,\|\, p_\theta(X_{t-1} \mid X_t)) && (391)\\ &\quad = \frac{\sigma_t^2}{2}\left\|(\tilde{\eta}_X(X_t) - b_t^\eta K_\theta(X_t, t)) - (\tilde{\eta}_X(X_t) - b_t^\eta K_t)\right\|_F^2 && (392)\\ &\quad = \frac{\sigma_t^2 (b_t^\eta)^2}{2}\|K_\theta(X_t, t) - K_t\|_F^2. && (393) \end{aligned}$$

Since

$$\sigma_t^2(b_t^\eta)^2 = \frac{\beta_t(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t} \cdot \frac{1-\bar{\alpha}_t}{\alpha_t(1-\bar{\alpha}_{t-1})^2} \tag{394}$$

$$= \frac{\beta_t}{\alpha_t(1-\bar{\alpha}_{t-1})}, \tag{395}$$

it follows that

$$D_{KL}(\tilde{q}(X_{t-1} \mid X_t, X_0) \,\|\, p_\theta(X_{t-1} \mid X_t)) = \frac{\beta_t}{2\alpha_t(1-\bar{\alpha}_{t-1})} \|K_\theta(X_t, t) - K_t\|_F^2. \tag{396}$$

Therefore, a natural training objective is

$$\tilde{L} = \mathbb{E}_q\Bigg[D_{KL}(q(X_T \mid X_0)\|p(X_T)) + \sum_{t>1} \frac{\beta_t}{2\alpha_t(1-\bar{\alpha}_{t-1})} \|K_\theta(X_t, t) - K_t\|_F^2 - \log p_\theta(X_0 \mid X_1)\Bigg]. \tag{397}$$

3.10 Inverse-Wishart Distribution

For the Inverse-Wishart $\mathcal{IW}_p(\nu, \Psi)$ distribution on the cone of symmetric positive definite matrices $\mathbb{S}_+^p$, we have the density

$$p_{\mathcal{IW}}(Z) \propto |Z|^{-(\nu+p+1)/2} \exp\Big(-\tfrac{1}{2}\mathrm{tr}(\Psi Z^{-1})\Big)\, \mathbf{1}_{\{Z \in \mathbb{S}_+^p\}}.$$

Let $\varepsilon_1, \varepsilon_2, \ldots, \varepsilon_T$ be a sequence of i.i.d. random matrices distributed as $\mathcal{IW}_p(\nu, \Psi)$. Assuming $\nu > p+1$, their mean is

$$M = \mathbb{E}[\varepsilon_t] = \frac{\Psi}{\nu - p - 1} \in \mathbb{S}_+^p.$$

For a fixed variance schedule $(\beta_t)_{t=1}^T$ with $\alpha_t = 1 - \beta_t$, we define the forward process

$$X_t = \sqrt{\alpha_t}\, X_{t-1} + \sqrt{\beta_t}\,(\varepsilon_t - M). \tag{398}$$

Then the conditional density takes the form

$$q(X_t \mid X_{t-1}) = \frac{1}{(\sqrt{\beta_t})^m} P_\varepsilon\left(\frac{X_t - \sqrt{\alpha_t}X_{t-1} + \sqrt{\beta_t}M}{\sqrt{\beta_t}}\right), \tag{399}$$

where $m := p(p+1)/2$ and P_ε is the density of the Inverse-Wishart noise. Ignoring the global normalizing constant depending only on ν, Ψ, we may write

$$q(X_t \mid X_{t-1}) = \frac{1}{(\sqrt{\beta_t})^m} |Z_t|^{-(\nu+p+1)/2} \exp\Big(-\tfrac{1}{2}\mathrm{tr}(\Psi Z_t^{-1})\Big)\, \mathbf{1}_{\{Z_t \in \mathbb{S}_+^p\}}, \tag{400}$$

where

$$Z_t = \frac{X_t - \sqrt{\alpha_t}X_{t-1} + \sqrt{\beta_t}M}{\sqrt{\beta_t}} = A_t - b_t X_{t-1}, \tag{401}$$

with

$$A_t = \frac{X_t + \sqrt{\beta_t}M}{\sqrt{\beta_t}}, \qquad b_t = \sqrt{\frac{\alpha_t}{\beta_t}}. \tag{402}$$

Therefore,

$$q(X_t \mid X_{t-1}) = \frac{1}{(\sqrt{\beta_t})^m} |A_t - b_t X_{t-1}|^{-(\nu+p+1)/2} \exp\Big(-\tfrac{1}{2}\mathrm{tr}\big(\Psi(A_t - b_t X_{t-1})^{-1}\big)\Big)\, \mathbf{1}_{\{A_t - b_t X_{t-1} \in \mathbb{S}_+^p\}}. \tag{403}$$

Thus, we write

$$q(X_t \mid X_{t-1}) = g_{IW}(X_t, X_{t-1}) \exp\big(\langle \eta_0(X_t), T_0(X_{t-1})\rangle\big), \tag{404}$$

where

$$g_{IW}(X_t, X_{t-1}) = \frac{1}{(\sqrt{\beta_t})^m} |A_t - b_t X_{t-1}|^{-(\nu+p+1)/2} \exp\Big(-\tfrac{1}{2}\mathrm{tr}\big(\Psi(A_t - b_t X_{t-1})^{-1}\big)\Big) \mathbf{1}_{\{A_t - b_t X_{t-1} \in \mathbb{S}_+^p\}}, \tag{405}$$

$$\eta_0(X_t) = 0, \tag{406}$$

$$T_0(X_{t-1}) = X_{t-1}, \tag{407}$$

$$A_0(\eta_0(X_t)) = 0. \tag{408}$$

Then, we define

$$\tilde{\eta}(X_t, X_0) := \eta_0(X_t) + \frac{\sqrt{\bar{\alpha}_{t-1}}}{1 - \bar{\alpha}_{t-1}} X_0 \tag{409}$$

$$= \frac{\sqrt{\bar{\alpha}_{t-1}}}{1 - \bar{\alpha}_{t-1}} X_0, \tag{410}$$

so that

$$\eta(X_t, X_0)^T T(X_{t-1}) = \big\langle \tilde{\eta}(X_t, X_0), X_{t-1} \big\rangle. \tag{411}$$

For the inverse Wishart distribution, we use the same log-partition form as above. Hence,

$$A(\eta(X_t, X_0)) = A(\tilde{\eta}(X_t, X_0)) \tag{412}$$

$$= \log \int_{\mathbb{S}^p} \exp\left(-\frac{\|X_{t-1}\|_F^2}{2(1 - \bar{\alpha}_{t-1})} \right) \exp\left(-\frac{1 - \beta_t}{2\beta_t} \|X_{t-1}\|_F^2 \right) \exp(\langle \tilde{\eta}(X_t, X_0), X_{t-1} \rangle)\, dX_{t-1}. \tag{413}$$

Equivalently,

$$A(\tilde{\eta}(X_t, X_0)) = \log \int_{\mathbb{S}^p} \exp\left(-\frac{1}{2} \left(\frac{1}{1 - \bar{\alpha}_{t-1}} + \frac{1 - \beta_t}{\beta_t} \right) \|X_{t-1}\|_F^2 + \langle \tilde{\eta}(X_t, X_0), X_{t-1} \rangle \right) dX_{t-1}. \tag{414}$$

Define

$$\sigma_t^2 := \frac{1}{\frac{1}{1 - \bar{\alpha}_{t-1}} + \frac{1 - \beta_t}{\beta_t}} = \frac{\beta_t (1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t}. \tag{415}$$

Using the Gaussian integral identity on the symmetric-matrix space, we obtain

$$A(\tilde{\eta}(X_t, X_0)) = \log \left((\sqrt{2\pi\sigma_t^2})^m \right) + \frac{\sigma_t^2}{2} \|\tilde{\eta}(X_t, X_0)\|_F^2, \tag{416}$$

where $m = \frac{p(p+1)}{2}$ is the dimension of $\mathbb{S}^p$.

Therefore,

$$D_{KL}(\tilde{q}(X_{t-1} \mid X_t, X_0) \,\|\, p_\theta(X_{t-1} \mid X_t)) \tag{417}$$

$$= A(\eta_\theta(X_t, t)) - A(\tilde{\eta}(X_t, X_0)) - \big\langle \eta_\theta(X_t, t) - \tilde{\eta}(X_t, X_0), \nabla A(\tilde{\eta}(X_t, X_0)) \big\rangle \tag{418}$$

$$= \frac{\sigma_t^2}{2} \left\| \eta_\theta(X_t, t) - \tilde{\eta}(X_t, X_0) \right\|_F^2. \tag{419}$$

Next, we express $\tilde{\eta}(X_t, X_0)$ in terms of the latent variable K_t. From the forward process,

$$X_t = \sqrt{\bar{\alpha}_t}\, X_0 + \sqrt{1 - \bar{\alpha}_t}\, K_t, \tag{420}$$

which implies

$$X_0 = \frac{X_t - \sqrt{1 - \bar{\alpha}_t}\, K_t}{\sqrt{\bar{\alpha}_t}}. \tag{421}$$

Substituting this into $\tilde{\eta}(X_t, X_0)$ gives

$$\tilde{\eta}(X_t, X_0) = \frac{\sqrt{\bar{\alpha}_{t-1}}}{1-\bar{\alpha}_{t-1}} \cdot \frac{X_t - \sqrt{1-\bar{\alpha}_t}\, K_t}{\sqrt{\bar{\alpha}_t}} \tag{422}$$

$$= \frac{1}{(1-\bar{\alpha}_{t-1})\sqrt{\alpha_t}}\, X_t - \frac{\sqrt{1-\bar{\alpha}_t}}{(1-\bar{\alpha}_{t-1})\sqrt{\alpha_t}}\, K_t. \tag{423}$$

Therefore, we can write

$$\tilde{\eta}(X_t, X_0) = \tilde{\eta}_X(X_t) - b_t^{\eta} K_t, \tag{424}$$

where

$$\tilde{\eta}_X(X_t) = \frac{1}{(1-\bar{\alpha}_{t-1})\sqrt{\alpha_t}}\, X_t, \tag{425}$$

$$b_t^{\eta} = \frac{\sqrt{1-\bar{\alpha}_t}}{(1-\bar{\alpha}_{t-1})\sqrt{\alpha_t}}. \tag{426}$$

We parameterize the model natural parameter in the same form:

$$\eta_\theta(X_t, t) = \tilde{\eta}_X(X_t) - b_t^{\eta} K_\theta(X_t, t), \tag{427}$$

or equivalently,

$$K_\theta(X_t, t) := \frac{\tilde{\eta}_X(X_t) - \eta_\theta(X_t, t)}{b_t^{\eta}}. \tag{428}$$

Substituting $\tilde{\eta}(X_t, X_0) = \tilde{\eta}_X(X_t) - b_t^{\eta} K_t$ and $\eta_\theta(X_t, t) = \tilde{\eta}_X(X_t) - b_t^{\eta} K_\theta(X_t, t)$ into (419), we obtain

$$D_{KL}(\tilde{q}(X_{t-1} \mid X_t, X_0) \,\|\, p_\theta(X_{t-1} \mid X_t)) \tag{429}$$

$$= \frac{\sigma_t^2}{2} \left\| (\tilde{\eta}_X(X_t) - b_t^{\eta} K_\theta(X_t, t)) - (\tilde{\eta}_X(X_t) - b_t^{\eta} K_t) \right\|_F^2 \tag{430}$$

$$= \frac{\sigma_t^2 (b_t^{\eta})^2}{2} \|K_\theta(X_t, t) - K_t\|_F^2. \tag{431}$$

Since

$$\sigma_t^2 (b_t^{\eta})^2 = \frac{\beta_t(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t} \cdot \frac{1-\bar{\alpha}_t}{\alpha_t(1-\bar{\alpha}_{t-1})^2} \tag{432}$$

$$= \frac{\beta_t}{\alpha_t(1-\bar{\alpha}_{t-1})}, \tag{433}$$

it follows that

$$D_{KL}(\tilde{q}(X_{t-1} \mid X_t, X_0) \,\|\, p_\theta(X_{t-1} \mid X_t)) = \frac{\beta_t}{2\alpha_t(1-\bar{\alpha}_{t-1})} \|K_\theta(X_t, t) - K_t\|_F^2. \tag{434}$$

Therefore, a natural training objective is

$$\tilde{L} = \mathbb{E}_q \left[D_{KL}(q(X_T \mid X_0) \| p(X_T)) + \sum_{t>1} \frac{\beta_t}{2\alpha_t(1-\bar{\alpha}_{t-1})} \|K_\theta(X_t, t) - K_t\|_F^2 - \log p_\theta(X_0 \mid X_1) \right]. \tag{435}$$